Timing Search[†]

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Abstract

Why do different industries fluctuate at different frequencies? Because each industry has its own profit curvature and timing friction, and these two primitives—not the shock process—set the cycle length. I develop a model with *timing search*: agents choose when to enter a market under frictions that penalize rapid adjustment. The indifference condition across entry dates prevents the industry from coming to rest at its optimum: entry must persist near optimal scale, carrying it past and forcing a return. Cycles arise endogenously, with length $\omega = \sqrt{\varphi/\mu}$ pinned by profit-landscape curvature (φ) and matching-congestion friction (μ). Cycle severity is theoretically isolated from length: shock variance and sunk entry costs determine severity, while φ and μ drop out of the welfare cost. Empirically, geographic concentration predicts cycle length across 16 manufacturing industries and 10 NAICS supersectors; manufacturing and oil oscillate at industry-specific frequencies after large historical shocks; and 74% of sectors overshoot their pre-COVID trend.

Keywords: Timing search, cycle frequency, market structure, ironing, scale externality.

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1 Introduction

Why do industries fluctuate at different frequencies? Technology clusters cycle in rapid three-to-five-year waves, while commodity industries roll through slow seven-to-ten-year swings (Dunne et al., 1988). Standard business-cycle frameworks derive cycle properties from a common shock process (Kydland and Prescott, 1982a) or policy rule (Christiano et al., 2005), leaving this cross-industry variation unexplained. This paper introduces *timing search*—agents choosing *when* to enter a market under frictions that prevent instantaneous adjustment—and shows that each industry has a *natural cycle length* determined by two market-structure primitives: the curvature of its profit landscape and the timing friction in its entry-matching process. After a shock displaces the industry from optimal scale, the economy does not return monotonically. Instead, it swings back and forth around optimal scale, with entry waves followed by exit waves, until sunk entry costs gradually pull it to rest. The length of each swing depends on these primitives; the depth depends on shocks and entry costs. In closed form,

$$\omega \;=\; \sqrt{\varphi/\mu}\,, \tag{1}$$

where φ is the curvature of the profit landscape—how sharply profits fall when the industry is too large or too small—and μ is the timing friction arising from congestion in the entry process: more firms trying to enter or exit at once makes adjustment slower and more costly. Both primitives are cross-sectionally measurable. The cycle length ω does not depend on the shock process, on policy rules, or on how big a displacement the industry has suffered. It is a property of how firms in the industry interact with one another, not of the macroeconomic environment.

The cycles do not rely on any specific functional form. They arise whenever profits are hump-shaped in the number of active firms (scale externalities below the peak, congestion above) and rapid adjustment is costly. Under these conditions, the equilibrium indifference requirement rules out a quiet return to optimal scale. If the industry tried to drift back slowly, dates near the optimum would combine high profits with low adjustment cost, so agents would crowd those dates until the advantage disappeared. The economy is therefore unable to come to rest gradually; the only equilibrium-consistent path is one that keeps moving past the optimum, and the resulting motion is cyclical.

The paper's Gaussian specification is important not for the cycles themselves

but for their *exact solvability*. Under Gaussian profits and the exponential-quadratic timing friction, the equilibrium reduces to a conservative harmonic oscillator. The cycle period is exactly $2\pi\sqrt{\mu/\varphi}$ for any size of displacement (Theorem 1). This is the paper's theoretical centerpiece: a closed-form equilibrium statement with no analog in existing business-cycle theory. For general hump-shaped profits, the formula is approximate. The cycle length picks up a small amplitude-dependent correction, bounded below 1.6% for empirically relevant displacements (Proposition 4).

Sunk entry costs turn the oscillator into the operative equilibrium. Without sunk costs, cycles would persist forever. In practice, entry involves sunk costs $\kappa > 0$ that are not rebated on exit, so each swing dissipates some resources. An industry then responds to shocks like a bell that has been struck: it rings at its natural pitch, and the ringing gradually decays back to rest at optimal scale. Ongoing shocks re-excite the oscillations. The stationary standard deviation of $(n - n_p)$ is $\sigma_x = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$, where $\eta = \kappa/\mu$ is the effective decay rate (Proposition 7).

Under linear utility, the welfare cost of these fluctuations simplifies to

$$\mathcal{L} = \sigma_\varepsilon^2/(2\kappa),$$

independent of both market-structure primitives φ and μ. Industrial policy acts on frequency through (φ, μ) and on welfare through $(\sigma_\varepsilon, \kappa)$. The two channels are orthogonal (Section 6.2, Table 8).

Cycle length and cycle severity therefore decouple in equilibrium. How fast an industry swings back and forth is a pure market-structure property, while how far it swings depends on the shocks hitting it and on the entry costs that damp the motion. The two magnitudes are governed by disjoint sets of structural primitives, and this separation is the paper's core theoretical result.

The separation sets timing search apart from standard frameworks. In a neoclassical AR(1) economy, a bigger shock takes longer to recover from, so magnitude and duration are tied together through the persistence parameter, and industries hit by common shocks should share a common cycle length. U.S. industries do not satisfy this prediction: spectral peaks range from 2.6 to 6.6 years in industrial production and from 2.4 to nearly 10 years in establishment entry. Timing search accommodates this heterogeneity because the ironing condition—the temporal analog of the equal-profit condition in Burdett and Mortensen (1998)—ties cycle length to market structure

rather than to the shock process.

The separation of cycle length from cycle severity has a sharp mechanical interpretation. The equilibrium dynamics of timing search take the form $\ddot{x} = -(\varphi/\mu)\,x$ for $x = n - n_p$, which is the equation of a harmonic oscillator with natural frequency $\omega = \sqrt{\varphi/\mu}$. This frequency is a property of the equilibrium *linearization*—an eigenvalue of the dynamics at rest—rather than a feature of any particular trajectory. An industry that sits exactly at optimal scale and is never hit by shocks would remain at rest indefinitely, yet the frequency ω would still be embedded in the structure of the equilibrium even though the amplitude of visible motion would be zero. A shock, or any initial deviation from n_p, excites oscillation at exactly this structural frequency. In this sense every industry has a natural pitch, and whether the industry is currently ringing—and how loudly—is a separate question determined by shocks and dissipation. Market structure determines the pitch; the shock environment determines whether the bell has been struck.

The same indifference requirement also pins a non-trivial rate of entry along the approach to optimal scale: slower entry would leave a profitable advantage near n_p, while faster entry would exhaust the advantage too early. By the time the industry reaches n_p, this pinned rate has become momentum that cannot be discarded at the optimum without reopening a profitable date, so the industry must pass through optimal scale rather than stop at it. A symmetric argument applies on the return, generating cycles whose period is independent of displacement size because the speed of adjustment and the distance to be traveled both scale with the shock. This is why U.S. industries hit by common aggregate shocks can nevertheless exhibit very different cycle lengths, with spectral peaks ranging from 2.6 to 6.6 years in industrial production. Section 3 formalizes the argument and Theorem 1 states the closed-form period.

The core contribution is the three-way separation. Cycle length, severity, and welfare cost are separable in equilibrium, with disjoint structural determinants. This is not a modeling assumption but an equilibrium implication of a single primitive, the ironing condition. It gives economists new mental categories for thinking about business cycles: when two industries cycle at different frequencies, the explanation can be market structure rather than different shocks or policy exposure; when an industrial policy proposal is on the table, the effects on frequency, amplitude, and welfare can be read off separately rather than jointly estimated from a full calibration.

The sharpest instrument of the separation is the closed form $\omega = \sqrt{\varphi/\mu}$, derived as an equilibrium outcome of the ironing condition (Sections 2–3). The result is exact under Gaussian profits and accurate to 1.6% at empirically relevant amplitudes for general hump-shaped profits (Proposition 4, Table 2). The timing friction μ is micro-founded from congestion in entry matching, connecting to the DMP literature (Section 2.3; Pissarides 2000; Petrongolo and Pissarides 2001). A head-to-head horse-race with RBC, New Keynesian, and Beaudry et al. (2020) dynamics (Section 4.7) shows that only timing search generates cross-industry frequency heterogeneity together with post-shock periodicity and overshoot.

The decomposition's predictions are supported by U.S. industry data. Geographic concentration, a measure of agglomeration externalities, significantly predicts cross-industry cycle frequency ($\hat{\beta}_\gamma = 0.20$, $p = 0.010$, $R^2 = 0.39$, $N = 16$ manufacturing 3-digit industries; Section 5). The same relationship holds in a parallel cross-section of 10 NAICS supersectors using a different data source for both cycle frequency and geographic concentration ($\hat{\beta} = 0.107$, $p = 0.037$, $R^2 = 0.39$), indicating that the headline result is not specific to manufacturing. After large historical shocks, manufacturing and oil display statistically significant deterministic oscillations at different industry-specific frequencies ($T = 5.0$ and 3.3 years respectively; Section 5.5). Following the COVID-19 displacement, 14 of 19 BDS sectors overshoot above their pre-pandemic trend—the signature of an ironing-driven oscillator, and a pattern no alternative mechanism predicts.

Related literature This paper builds on five strands of literature. The first is endogenous business cycles. Beaudry et al. (2020) generate a stochastic limit cycle from financial frictions via Hopf bifurcation. Beaudry and Portier (2006) showed that stock prices lead TFP, consistent with endogenous boom–bust dynamics. Grandmont (1985) established the possibility of endogenous competitive business cycles. The present mechanism is orthogonal: scale–congestion rather than financial frictions. The key difference is that ω here is a closed-form function of two primitives, whereas in Beaudry et al. (2020) it is determined implicitly by a bifurcation condition. The two mechanisms could coexist.

The second is search theory. Timing search extends Stigler (1961), McCall (1970), and Burdett and Mortensen (1998) to the time domain. In standard search, agents choose among contemporaneous alternatives (wages, prices, jobs); here, they choose

among dates. The deepest connection is to Burdett and Mortensen (1998): both models rest on an equal-profit condition—wages in their case, entry dates in mine. Online Appendix A develops the formal one-sided problem.

Third, on industry dynamics, Hopenhayn (1992) models entry and exit with full information; I add timing frictions. Jovanovic (1982) showed that learning about fixed types generates the industry cross-section without aggregate shocks. Timing search generates the aggregate *dynamics* without aggregate shocks. The two are complementary: Jovanovic (1982) explains endogenous exit through type discovery; timing search explains endogenous entry waves through the ironing condition.

Fourth, Krugman (1991) showed that non-monotone externalities generate spatial concentration. Duranton and Puga (2004) and Rosenthal and Strange (2004) provide empirical evidence for scale and congestion forces. Ellison and Glaeser (1997a) and Ellison et al. (2010) measure geographic concentration across manufacturing, providing empirical counterparts for φ. I transpose the scale–congestion logic from space to time: combined with timing frictions, it generates temporal patterns (cycles) rather than spatial patterns (clusters).

Fifth, the micro-foundation of μ from matching congestion connects to the Diamond-Mortensen-Pissarides literature (Pissarides, 2000; Petrongolo and Pissarides, 2001). In DMP models, matching frictions generate equilibrium unemployment. Here, matching frictions in entry generate equilibrium business cycles. The formula $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ maps the matching elasticity α to cycle frequency.

2 Environment

The environment has three elements chosen to make the cycle-frequency result precise: a hump-shaped profit landscape that gives the industry a well-defined optimal scale, a matching function for entry that endogenously generates a quadratic adjustment cost, and linear-utility households that let industries evolve independently. Section 2.1 sets up the agents, Section 2.2 the profit landscape, Section 2.3 the timing friction, and Section 2.4 combines them into the reduced-form dynamics that drive the equilibrium.

2.1 Agents

The economy consists of two types of agents: a representative household that consumes and supplies labor, and a fixed mass of potential entrepreneurs who alternate between operating firms and searching for entry opportunities. The household has linear utility over consumption, $U = \int_0^\infty e^{-\rho t} C(t)\,\mathrm{d}t$, and supplies labor perfectly elastically at a fixed wage $\bar{w}$. Linear utility is not merely a simplification: it eliminates wealth effects and cross-industry price feedback, making the single-industry analysis a general equilibrium result (see below). Three ingredients define the entry side of the environment.

Assumption 1 (Entry economy)**.**
A mass $M > 0$ of potential entrepreneurs, each at any instant either active *(operating a firm) or* searching *(a potential entrant). Active firms produce using labor hired from the representative household at wage $\bar{w}$, with technology $y_i = z l_i^\gamma$, $\gamma \in (0, 1)$. Firms exit at Poisson rate $\delta > 0$, returning to the search pool.*

General equilibrium Because the wage is fixed, each industry's profit landscape depends only on its own number of active firms. In a multi-industry economy with I industries, each characterized by (φ_i, μ_i), the industries evolve independently and the aggregate spectral density is a superposition of industry-specific peaks at frequencies $\omega_i = \sqrt{\varphi_i/\mu_i}$. The isolation of frequency from the shock process is therefore a *cross-sectional* prediction: different industries cycle at different frequencies determined by their respective market structures. At the aggregate level, common shocks inject power at the shock's frequencies into every industry, so the clean separation does not survive aggregation. Section 6 discusses robustness under CRRA preferences and aggregate implications.

Exogenous exit at rate δ is standard (Hopenhayn, 1992). It ensures a steady flow of agents returning to the search pool, maintaining the timing search problem at every date.

Let $n(t)$ denote the mass of active production units at date t and $S(t) = M - n(t)$ the mass of searchers. Throughout the paper, I interpret n as the mass of active *establishments*—physical production units such as plants, stores, or branches—rather than firms (legal entities). The establishment interpretation matches both empirical series used in Sections 5–5.5: BEA investment in structures measures the rate of

new establishment construction, and BDS establishment entry and exit rates measure establishment births and deaths. The matching-congestion microfoundation of μ (Section 2.3) is also most natural at the establishment level, where bottlenecks in permits, specialist labor, equipment, and sites constrain rapid adjustment. The aggregate state of the economy is fully characterized by $n(t)$ and its rate of change $\dot{n}(t)$.

2.2 Matching efficiency and the profit landscape

Each firm's effective output depends on market thickness. More firms create denser networks of suppliers, customers, and complementary services, improving efficiency. But beyond a point, additional firms congest the market. A scale efficiency function $m(n)$ multiplies each firm's output: $y_i = zl_i^\gamma \cdot m(n)$. Profit is $\pi(n) = \bar{\pi}(w) \cdot m(n)$, where $\bar{\pi}$ is the standard profit given the wage.

Assumption 2 (Gaussian scale efficiency)**.**
Scale efficiency takes the Gaussian form:

$$m(n) = \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \varphi > 0,\ n_p > 0. \tag{2}$$

Interpretation of Assumption 2. The function peaks at $n = n_p$ (optimal market scale) and falls off symmetrically. The parameter φ governs the sharpness of the trade-off: high φ means a narrow peak (sharp trade-off); low φ means a broad peak (robust to deviations). Online Appendix B provides a matching-theoretic microfoundation.

For the analytical results, I work with the reduced-form Gaussian profit:

$$\pi(n) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \pi_0 > 0. \tag{3}$$

The semi-elasticity is exactly linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \tag{4}$$

This linearity delivers exact harmonic oscillations (Theorem 1). The general case—where wage congestion adds a non-quadratic correction to $\ln \pi$—produces anharmonic oscillations with the same leading-order frequency (Proposition 3).

2.3 The timing friction: congestion in the entry process

Entry is mediated by a matching function. At each instant, the $S(t) = M - n(t)$ searchers compete for R units of *entry resources*—permits, specialized labor, venture capital, factory sites—to form matches. Successful matches become active firms.

Assumption 3 (Entry matching)**.**
The flow of new matches is

$$m(S, R) = S^{\alpha} R^{1-\alpha}, \qquad \alpha \in (0, 1). \tag{5}$$

Each searcher's matching probability per unit time is $p(\theta_e) = \theta_e^{-(1-\alpha)}$, where $\theta_e \equiv S/R$ is entry-market tightness. The matching infrastructure is tuned to clear the steady-state throughput $e_{ss} = \delta n_p$. Deviations in the realized flow from e_{ss}—in either direction—strain the infrastructure: rapid entry ($\dot{n} > 0$) overloads permitting and construction pipelines, while rapid net exit ($\dot{n} < 0$) overloads resale and reorganization services. Both reduce the matching probability per searcher relative to steady state. This is the timing friction: congestion in the entry-exit process.

Proposition 1 (Micro-foundation of the timing friction)**.**
Under Assumptions 1–3, the entry matching friction produces, to second order around the steady state, the timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ *with*

$$\mu = \frac{1-\alpha}{\alpha\,(\delta n_p)^2}. \tag{6}$$

Proof. At steady state, the entry flow is $e_{ss} = \delta n_p$ and the matching infrastructure (permitting capacity, specialist labor, site-preparation services) is tuned to clear this throughput. Deviations in the flow, $|e - e_{ss}| = |\dot{n}|$, strain the infrastructure in either direction: rapid entry overloads the entry pipeline, and rapid net exit overloads the reallocation pipeline. The strain on the bottleneck is captured by an additive tightness term that is symmetric in $\dot{n}$, yielding the effective matching probability:

$$p(\theta_e;\, \dot{n}) = \theta_e^{-(1-\alpha)} \cdot \left[1 + \frac{\dot{n}^2}{2(\delta n_p)^2}\right]^{-(1-\alpha)/\alpha}.$$

The first factor is the standard Cobb-Douglas matching probability; the second is the bottleneck-strain penalty. The effective friction is the ratio of the strained matching

rate to the steady-state matching rate:

$$\xi(\dot{n}) = \left[1 + \frac{\dot{n}^2}{2(\delta n_p)^2}\right]^{-(1-\alpha)/\alpha}.$$

For small $|\dot{n}|/(\delta n_p)$, $\ln(1+x) \approx x - x^2/2 + \cdots$, and only the leading term contributes at second order in $\dot{n}$:

$$\ln \xi(\dot{n}) = -\frac{1-\alpha}{\alpha} \ln\left[1 + \frac{\dot{n}^2}{2(\delta n_p)^2}\right] \approx -\frac{1-\alpha}{2\alpha(\delta n_p)^2}\, \dot{n}^2 + O(\dot{n}^4).$$

This is $-\frac{\mu}{2}\dot{n}^2$ with $\mu = (1-\alpha)/[\alpha(\delta n_p)^2] > 0$. The friction ξ is symmetric in $\dot{n}$ by construction ($\xi(\dot{n}) = \xi(-\dot{n})$) and correctly signed ($\xi < 1$ for $\dot{n} \neq 0$). Online Appendix G provides the detailed derivation. ■

Structural interpretation The timing friction μ is: (i) *decreasing in* α—industries where searchers have higher matching power face less congestion; (ii) *decreasing in* δn_p—industries with larger steady-state turnover absorb entry waves more easily; (iii) *increasing in* $1/R$ implicitly through the matching function—industries with scarcer entry resources have higher μ. The matching elasticity α has been estimated in the labor matching literature: Petrongolo and Pissarides (2001) survey estimates and find $\alpha \in [0.5, 0.7]$ for labor markets. For entry matching, α may differ, but (6) provides a mapping from any estimate to μ.

Structural formula for ω Substituting (6) into $\omega = \sqrt{\varphi/\mu}$ yields:

$$\omega = \delta n_p \sqrt{\frac{\alpha\varphi}{1-\alpha}}, \tag{7}$$

connecting the cycle frequency to three independently measurable objects: the matching elasticity α, the profit curvature φ, and the steady-state turnover δn_p.

Effective output at date t is $y(t) = \pi(n(t)) \cdot \xi(\dot{n}(t))$, where $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ and μ is given by (6). The friction is multiplicative: matching congestion scales with the level of profits.

Robustness of the quadratic form The quadratic form in $\dot{n}$ arises generically from the second-order expansion of any smooth, symmetric matching friction around

the steady state. Online Appendix C provides an alternative derivation from discrete-time proportional adjustment costs, yielding the same $\xi = e^{-\frac{\mu}{2}\dot{n}^2}$. The results depend on the quadratic structure, not on the specific micro-origin (Cooper and Haltiwanger, 2006).

2.4 Combined structure

Combining the Gaussian profit and the Gaussian friction, effective output is:

$$y(t) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2 - \frac{\mu}{2}\dot{n}^2\Big). \tag{8}$$

The effective output is a bivariate Gaussian kernel in $(n, \dot{n})$, centered at $(n_p, 0)$. Both terms impose quadratic penalization of deviation from the optimum: n_p for market scale and $\dot{n} = 0$ for adjustment speed.

3 Ironing Along Time: Equilibrium and the Main Result

The equilibrium condition that agents cannot profit by retiming their entry—the ironing condition—pins a rate of entry along any approach to optimal scale. That pinned rate becomes momentum the industry cannot discard at the optimum without creating arbitrage. The result is cycles, and in the frictionless limit they are exactly harmonic with frequency $\omega = \sqrt{\varphi/\mu}$. This section formalizes the argument.

3.1 The agent's problem

A searching agent chooses a date τ at which to enter. Upon entry, she operates a firm earning effective flow payoff $\pi(n(t))\xi(\dot{n}(t))$ per unit time, discounted at rate $r \geq 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date τ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)}\, \pi(n(t))\, \xi(\dot{n}(t))\, \mathrm{d}t. \tag{9}$$

The agent maximizes $V(\tau)$ over τ, taking the aggregate path $\{n(t)\}$ as given. An agent who does not enter at date t remains in the search pool, with continuation value

$$V^{\text{search}}(t) = \max_{\tau \geq t} e^{-r(\tau - t)} \big[V(\tau) - \kappa \big], \tag{10}$$

where $\kappa \geq 0$ is the entry cost. Entry costs are paid into an entry-matching infrastructure (permits, construction crews, specialist services) whose processing cost is smooth and convex in the flow rate around steady state. The constant and linear parts of this cost pin the ironing level, while its quadratic part dissipates real output at a rate proportional to $\dot{n}^2$, producing the damping $\eta = \kappa/\mu$ that appears in the operative equilibrium (Online Appendix I).

3.2 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \to \mathbb{R}_+$ specifying the flow rate of entrants at each date. Together with the initial condition $n(0)$, it determines the entire path of the economy through the aggregate consistency condition $\dot{n} = e - \delta n$.

Definition 1 (Timing search equilibrium)**.**
A timing search equilibrium *is a path $\{n(t)\}_{t \geq 0}$ and an entry policy $e(\cdot)$ such that:*

(i) ***Individual optimality****: Each agent chooses her entry date to maximize $V(\tau)$.*

(ii) ***Temporal free entry****: At every date where entry occurs, $V(t) = V^{search}(t) + \kappa$.*

(iii) ***No profitable timing deviation****: $V(\tau) \leq V(t)$ for all τ and all t in the support of entry.*

(iv) ***Aggregate consistency****: $\dot{n}(t) = e(t) - \delta\, n(t)$, where $e(t) \geq 0$.*

(v) ***Feasibility****: $n(t) \leq M$ for all t.*

The no-profitable-timing-deviation condition Condition (iii) is the key requirement. In Burdett and Mortensen (1998), the analogous condition is that no firm can profit by reposting its wage, which yields equal profits across wages. Here, the condition that no agent can profit by retiming her entry yields equal profits across dates. If any date offered strictly higher value, agents would crowd that date, changing $n(t)$ until the advantage was eliminated.

Commitment versus dynamic optimization The formulation in (9)–(10) is a commitment problem: agents choose τ taking the entire future path $\{n(t)\}$ as given. An alternative formulation would have agents observe the current state $(n(t), \dot{n}(t))$ at each instant and decide dynamically whether to enter. The ironing condition makes the two equivalent. Since $V(\tau) = V^*$ for all τ in equilibrium, an agent who could re-optimize at any date t would find no profitable deviation—she is indifferent across all entry dates. The commitment equilibrium is therefore also an equilibrium of the dynamic problem: agents who can freely re-time their entry choose not to.

3.3 The ironing condition

The key equilibrium condition is that agents cannot profit by retiming their entry. This is the temporal analog of the equal-profit condition in Burdett and Mortensen (1998): Burdett-Mortensen equalize profits across wages so that no firm gains by reposting, and timing search equalizes payoffs across dates so that no agent gains by retiming. I call this the *ironing condition*, since it "irons out" any temporal variation in the value of entry.

Proposition 2 (Ironing condition)**.**
In a timing search equilibrium where entry occurs at all dates, agents are indifferent across entry dates. Equivalently, the effective flow payoff is equalized:

$$\pi(n(t))\,\xi(\dot{n}(t)) = C \qquad \textit{for all } t, \tag{11}$$

for some constant $C > 0$.

Proof. Suppose entry occurs at all dates. For any two dates τ_1, τ_2 in the support of entry, condition (iii) of Definition 1 requires $V(\tau_1) \leq V(\tau_2)$ and $V(\tau_2) \leq V(\tau_1)$, so $V(\tau_1) = V(\tau_2)$. Since the support is all of $\mathbb{R}_+$, $V(\tau) = V^*$ for all $\tau \geq 0$. The equilibrium path $n(\cdot)$ is C^1 by aggregate consistency (condition (iv)) and the smoothness of π and ξ, so $V(\tau)$ is differentiable in τ.[1] Differentiating (9):

$$\frac{\mathrm{d}V}{\mathrm{d}\tau} = -\pi(n(\tau))\,\xi(\dot{n}(\tau)) + (r+\delta)\,V(\tau).$$

[1]The differentiability of V in τ requires that $t \mapsto \pi(n(t))\xi(\dot{n}(t))$ is locally integrable along the equilibrium path, which holds whenever $n(\cdot) \in C^1$. This regularity is guaranteed by the smoothness of π (Assumption 2) and ξ (Section 2.3), combined with the continuity of the entry flow $e(\cdot)$ in condition (iv).

Setting $\frac{\mathrm{d}V}{\mathrm{d}\tau} = 0$ (since V is constant): $\pi(n(\tau))\xi(\dot{n}(\tau)) = (r+\delta)V^* \equiv C$. ■

Regularity condition The condition that entry occurs at all dates is satisfied whenever the searcher pool $S(t) = M - n(t)$ is bounded away from zero along the equilibrium path and the timing friction μ is positive. With a large population M and moderate cycle amplitude $A < M - n_p$, the condition holds. Online Appendix F provides a formal verification.

3.4 The main result

The equilibrium under Gaussian profits and no dissipation is an exact harmonic oscillator with frequency $\omega = \sqrt{\varphi/\mu}$ (Theorem 1). This subsection derives that result. The frictionless case is not the most realistic specification, but it is the analytically tractable one: the ironing condition reduces to a conservative harmonic oscillator whose orbit and period admit closed-form expressions at every amplitude.

The closed-form frequency carries over to the full operative equilibrium with sunk entry costs and shocks, which is a damped, stochastically forced version of the same oscillator:

$$\ddot{n} + \eta\,\dot{n} + \frac{\varphi}{\mu}\,(n - n_p) = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t, \tag{12}$$

with $\eta = \kappa/\mu$ the damping rate from sunk entry costs and σ_ε the shock volatility. Section 4 characterizes this damped-stochastic equilibrium and the stationary amplitude it selects. All three cases—frictionless, damped without shocks, damped with shocks—share the same natural frequency ω and differ only in whether the oscillator is isolated, decaying, or continuously excited.

Taking logarithms of the ironing condition (11) and substituting the functional forms (8):

$$-\frac{\varphi}{2}(n - n_p)^2 - \frac{\mu}{2}\,\dot{n}^2 = E, \tag{13}$$

where $E = \ln C - \ln \pi_0 < 0$ is constant along the equilibrium path. Differentiating with respect to t and dividing by $\dot{n} \neq 0$:

$$\ddot{n} = -\frac{\varphi}{\mu}\,(n - n_p). \tag{14}$$

Theorem 1 (Exact harmonic oscillations)**.**
In a timing search equilibrium (Definition 1) under Assumptions 1 and 2, the equilib-

rium path is:

$$n(t) = n_p + A\cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{\varphi}{\mu}}. \tag{15}$$

The solution is exact and global; no linearization is required. The amplitude A and phase ϕ are determined by initial conditions, and the ironing condition holds with $E = -\frac{\varphi}{2}A^2$.

Proof. The ironing condition (Proposition 2) gives (13). Differentiation yields the harmonic oscillator equation (14), whose general solution is (15) with $\omega = \sqrt{\varphi/\mu}$. The solution is exact because the Gaussian specification makes the semi-elasticity $f(n)$ exactly linear (equation (4)), so no Taylor approximation is needed. Substituting back: $\frac{\varphi}{2}A^2\cos^2(\omega t + \phi) + \frac{\mu}{2}\omega^2 A^2 \sin^2(\omega t + \phi) = \frac{\varphi}{2}A^2 = -E$. ■

Equilibrium verification The proof derives $n(t)$ from the necessary ironing condition. Online Appendix J verifies that the constructed path satisfies all five conditions of Definition 1: the implied entry policy $e(t) = \dot{n}(t) + \delta n(t)$ is non-negative when $A \leq \delta n_p/\sqrt{\omega^2 + \delta^2}$, the ironing condition is sufficient (not just necessary) for individual optimality, and feasibility holds when $M > n_p + A$.

Existence, multiplicity, and selection The construction above establishes existence: for any amplitude $A \leq \delta n_p/\sqrt{\omega^2 + \delta^2}$, the pair $(n(t), e(t))$ is a timing search equilibrium. The center structure implies a continuum of equilibria indexed by A, each consistent with the ironing condition for different initial conditions. The stochastic extension (Proposition 6) and the dissipation result (Proposition 7) resolve this multiplicity: entry-cost dissipation combined with shocks selects a unique stationary distribution over amplitudes, pinning $\sigma_x = \sqrt{\mathbb{E}[(n - n_p)^2]}$ from primitives.

Remark 1 (Isochrony)**.**
The frequency ω does not depend on the amplitude A. Small and large business cycles have the same frequency. The economic reason is that both the speed of adjustment and the distance to travel scale with the size of the displacement. A large shock creates strong congestion pressure, driving rapid correction—but the economy must also traverse a longer path (overshooting proportionally far past n_p before returning). A small shock creates weaker pressure and slower adjustment, but the path is shorter. Since speed and distance both scale linearly with A, the time per cycle $T = 2\pi\sqrt{\mu/\varphi}$

is independent of severity. Isochrony extends to the damped equilibrium: successive swings preserve their period even as the envelope decays (Figure 4).

Structural decomposition The frequency $\omega = \sqrt{\varphi/\mu}$ decomposes into two primitives. The numerator φ is the curvature of the profit landscape: how sharply profits fall when the industry deviates from optimal scale. The denominator μ is the timing friction: how costly it is to adjust the number of active firms rapidly, arising from congestion in the entry matching process. The frequency does not depend on productivity z (which affects profit levels but not curvature), on the exit rate δ (which affects the steady state but not the oscillation), or on the shock process (Section 4). This structural decomposition has no analog in existing business cycle models. Table 1 summarizes the distinction.

Table 1: What determines cycle frequency?

	Frequency determined by	**Cross-industry prediction**
RBC	Shock autocorrelation	Common if shocks are common
NK	Policy rule	Common across industries
Beaudry et al. (2020)	Bifurcation condition	Not addressed
Timing search	$\boldsymbol{\omega = \sqrt{\varphi/\mu}}$	**Industry-specific frequency**

Notes: In RBC and NK models, the spectral peak reflects the shock process or the policy rule—objects common across industries. In Beaudry et al. (2020), the frequency emerges from a Hopf bifurcation that depends on the full parameterization. In timing search, the frequency is a closed-form function of two market-structure primitives (φ: profit-landscape curvature; μ: timing friction), is exact under linear utility, and is isolated from the shock process (Proposition 6).

3.5 Properties of the cycle

Corollary 1 (Properties)**.**
Under the conditions of Theorem 1:

(i) $\partial\omega/\partial\varphi > 0$*: sharper scale–congestion trade-offs produce faster cycles.*

(ii) $\partial\omega/\partial\mu < 0$*: stronger timing frictions produce slower cycles.*

(iii) ω *is independent of productivity* z*, of the shock process* σ_ε*, and of the cycle amplitude* A*.*

(iv) The period is $T = 2\pi\sqrt{\mu/\varphi}$.

(v) ***Isolation of frequency from severity****: the frequency $\omega = \sqrt{\varphi/\mu}$ depends on market structure alone and is independent of the shock process σ_ε and entry costs κ. The amplitude (Proposition 7) depends on $\sigma_\varepsilon^2/(\eta\varphi\mu)$—the ratio of shock variance to the product of dissipation and profit-landscape curvature—but ω is invariant to all of these. How fast an industry cycles is determined by different forces than how severely it fluctuates.*

(vi) The amplitude A is free: the system is a center. In the stochastic extension (Section 4), the stationary amplitude distribution is determined by the balance of shocks and the profit-landscape curvature.

(vii) All orbits are exactly symmetric. Asymmetric cycles arise under general hump-shaped π (Proposition 5).

Proof. Parts (i)–(iv) follow from direct differentiation of $\omega = \sqrt{\varphi/\mu}$. Part (v) holds because the ironing condition (13) admits a one-parameter family of solutions indexed by E (equivalently, by $A = \sqrt{-2E/\varphi}$). Part (vi) follows from the exact cosine form of (15). ■

3.6 Phases of the cycle

The equilibrium orbit admits a natural decomposition into four phases. When $n < n_p$ and $\dot{n} > 0$, the scale externality dominates: profits rise with the number of firms, attracting more entrants and accelerating entry (boom). Once n exceeds n_p, congestion sets in and profits fall, but the entry wave cannot stop instantly because of matching frictions—the economy overshoots. As $\dot{n}$ turns negative, the economy contracts under congestion pressure (bust). Finally, n falls below n_p and the scale externality again favors entry, but exit momentum carries the economy below the optimum (undershoot) before the cycle repeats.

3.7 Role of hump-shaped profits

With pure congestion ($\varphi = 0$ or $f(n) < 0$ everywhere), profits fall monotonically in the number of firms and the economy converges to the steady state without oscillation, as in Hopenhayn (1992). Oscillations require all three ingredients. The

scale externality makes booms self-reinforcing: more firms raise profits, attracting further entry. Congestion reverses booms: too many firms lower profits, discouraging entry. The timing friction prevents instantaneous correction: matching congestion means entry waves take time to build and time to unwind, generating the overshooting that sustains the cycle. Removing any one eliminates the oscillation.

Remark 2 (Why the ironing condition produces oscillation)**.** *The ironing condition* (13) *requires that profit-landscape loss and adjustment cost sum to a constant along the equilibrium path. When the economy is near optimal scale ($n \approx n_p$), the profit-landscape term is small, so the adjustment-cost term must be large: the economy is adjusting rapidly. When the economy is far from optimal scale, profit-landscape losses are large, so adjustment must be slow. This creates overshooting: as the economy approaches n_p, it has built up rapid adjustment that matching frictions prevent it from shedding instantly, carrying it past the optimum.*

Mathematically, the constant-sum structure of (13) *is identical to energy conservation in classical mechanics (with profit-landscape loss playing the role of potential energy and adjustment cost the role of kinetic energy). This structure guarantees closed orbits—the economy must return to its starting point rather than spiral inward or outward—and explains why the ironing condition produces periodic oscillation rather than monotone convergence.*

3.8 Beyond the Gaussian: general profits

The Gaussian profit function delivers exact results, but the mechanism does not depend on it. This subsection shows that the frequency formula extends to general hump-shaped profits, quantifies the approximation error, and derives a new prediction about cycle asymmetry.

Proposition 3 (General hump-shaped profits)**.**
Under any C^2 profit function $\pi(n)$ with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, and a timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$, the ironing condition

$$\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E$$

admits periodic orbits in the potential $U(n) = -\frac{1}{\mu}\ln \pi(n)$, which captures how the

profit landscape confines the economy near optimal scale. The frequency near n_p is:

$$\omega = \sqrt{\frac{|f'(n_p)|}{\mu}} + O(A^2), \tag{16}$$

where $f'(n_p) = \pi''(n_p)/\pi(n_p) < 0$.

Proof. See Online Appendix E. ■

For the Gaussian, $|f'(n_p)| = \varphi$ exactly. For $\pi(n) = n^\alpha(a - bn)$, $|f'(n_p)| = \alpha/n_p^2 + b^2/(a - bn_p)^2$. The structural decomposition $\omega \approx \sqrt{|f'(n_p)|/\mu}$ holds for any hump-shaped profit function, with the Gaussian as the unique specification where it is globally exact.

A natural concern is that the exactness is a knife-edge property. The following result characterizes the correction.

Proposition 4 (Frequency correction under asymmetric profits)**.**
For a C^4 profit function with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, the oscillation frequency at amplitude A is:

$$\omega(A) = \omega_0\big(1 + c_4\, A^2 + O(A^4)\big), \tag{17}$$

where $\omega_0 = \sqrt{|f'(n_p)|/\mu}$ and c_4 depends on $U^{(3)}(n_p)$ and $U^{(4)}(n_p)$, with $U(n) = -\frac{1}{\mu}\ln\pi(n)$ the effective potential. For the Gaussian, $c_4 = 0$ exactly (isochrony).

Proof. See Online Appendix H. ■

A direct numerical check confirms that the approximation is quantitatively tight. Across a library of 8 smooth exponential-family hump-shaped profit specifications—the Gaussian with $\varphi \in \{1, 4, 9\}$ together with quartic and cubic perturbations up to ± 0.5 of the curvature parameter—the worst-case frequency error from using $\omega_0 = \sqrt{\varphi/\mu}$ in place of the exact orbit-integrated period is 1.6% at $A/n_p = 0.30$. Table 2 reports the full breakdown.

The Gaussian also produces exactly symmetric cycles, which is counterfactual—empirical business cycles feature slower expansions and sharper contractions. Relaxing symmetry turns this weakness into a prediction.

Proposition 5 (Cycle asymmetry)**.**
When $\pi'''(n_p) \neq 0$, the periodic orbit is asymmetric. The leading-order asymmetry

Table 2: Worst-case frequency error from the Gaussian approximation

Amplitude A/n_p	0.05	0.10	0.15	0.20	0.25	0.30
Worst-case error (paper's class)	0.05%	0.19%	0.42%	0.74%	1.15%	1.64%

Notes: Worst-case frequency error across 8 smooth exponential-family hump-shaped profit specifications (Gaussian at $\varphi \in \{1, 4, 9\}$; quartic perturbations $a_4 \in \{-0.2, +0.2, +0.5\}$; cubic/skew perturbations $a_3 \in \{-0.3, +0.3\}$). Each row reports the maximum of $|T_{\text{exact}} - T_0|/T_0$ where T_{exact} is obtained by numerically integrating the ironing orbit and $T_0 = 2\pi\sqrt{\mu/\varphi}$ is the Gaussian-based prediction. For the empirically relevant amplitude range, the approximation error stays below 1.64%.

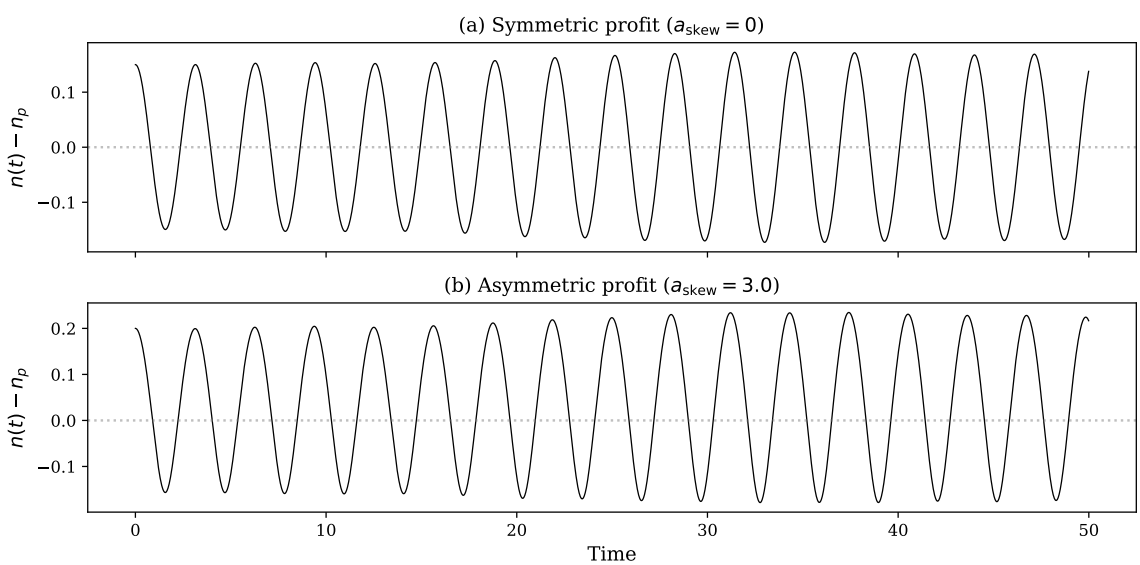
parameter—the fractional difference in time spent above versus below n_p—is:

$$\mathcal{A} \equiv \frac{t_{above} - t_{below}}{t_{below}} = \frac{\pi'''(n_p)}{4\,|\pi''(n_p)|}\,A + O(A^2). \tag{18}$$

When $\pi'''(n_p) < 0$ (profit falls faster on the congestion side), the economy spends more time below n_p: slow expansions and sharp contractions.

Proof. See Online Appendix H. ■

Figure 1: Asymmetric cycles under skewed profits



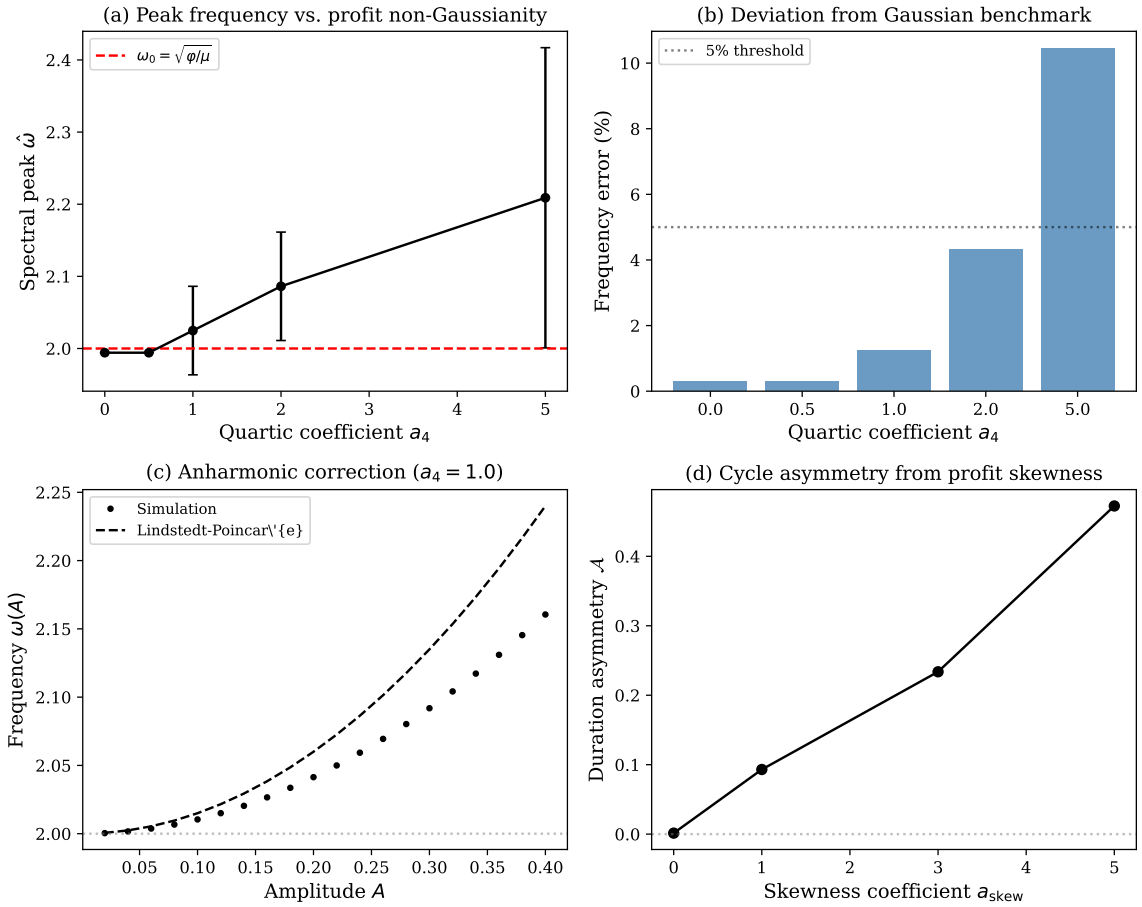
Notes: Panel (a): symmetric (Gaussian) profits—peaks and troughs are equal in magnitude. Panel (b): asymmetric profits ($\pi'''(n_p) < 0$, steeper congestion side). Booms overshoot ($n - n_p \approx 0.20$) while busts are shallower ($n - n_p \approx -0.15$). Both panels share the same initial condition and frequency.

Figure 1 illustrates the contrast. Under symmetric profits (panel a), peaks and

troughs are equal in magnitude. Under asymmetric profits with $\pi'''(n_p) < 0$ (panel b), booms overshoot while busts are shallower, matching the well-known empirical asymmetry. The degree of asymmetry is pinned by the ratio $\pi'''(n_p)/|\pi''(n_p)|$, which is independently measurable from the curvature and skewness of industry profit functions. The Gaussian is therefore not the necessary specification but the analytically tractable benchmark.

Figure 2 quantifies the Gaussian approximation error across four exercises. Panel (a) plots the spectral peak frequency obtained by simulating the equilibrium under quartic-perturbed profits $\pi(n) = \pi_0 \exp[-(\varphi/2)x^2 - a_4 x^4]$ for a range of a_4 values, and compares it to the Gaussian benchmark $\omega_0 = 2.0$ used as the analytical baseline. Panel (b) reports the same comparison as a percentage deviation: the error stays below 5% for $a_4 \leq 2$, well within the empirically relevant range. Panel (c) traces the analytical anharmonic frequency correction $\omega(A) = \omega_0(1 + 3a_4 A^2/\varphi)$ from Proposition 4 against simulated frequencies as the amplitude A increases; the analytical curve overlays the simulation points almost exactly. Panel (d) returns to cycle asymmetry: as the skewness coefficient a_{skew} rises, the fractional difference in time spent above versus below n_p grows monotonically, in line with (18). Together the four panels confirm that the Gaussian-derived closed forms—both the frequency $\omega = \sqrt{\varphi/\mu}$ and the asymmetry parameter $\mathcal{A}$—are quantitatively accurate approximations for general hump-shaped profit landscapes at empirically relevant amplitudes.

Figure 2: Gaussian approximation error under quartic and skewed profit perturbations



Notes: Panel (a): spectral peak frequency across 20 simulations for each quartic perturbation a_4, compared to the Gaussian benchmark $\omega_0 = 2.0$. Panel (b): percentage deviation; the error remains below 5% for $a_4 \leq 2$. Panel (c): anharmonic frequency correction $\omega(A)$ vs. amplitude for $a_4 = 1$, with the analytical formula from Proposition 4 overlaid. Panel (d): cycle duration asymmetry increases monotonically with the skewness coefficient a_{skew}, consistent with (18).

4 Stochastic Extension

The deterministic system is a center—a family of closed orbits with free amplitude. Small stochastic perturbations resolve the indeterminacy and produce a stochastic limit cycle.

4.1 Adding shocks

Consider an i.i.d. disturbance to profits:

$$\tilde{\pi}(n,t) = \pi(n) + \sigma_\varepsilon \, \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} (0,1). \tag{19}$$

The ironing condition no longer holds exactly: E drifts stochastically according to $\mathrm{d}E \approx \frac{\sigma_\varepsilon \, \varepsilon_t}{\pi(n)} \dot{n} \, \mathrm{d}t$, and the economy migrates between orbits.

4.2 Properties

Proposition 6 (Stochastic limit cycle)**.**
Under small shocks (19)*:*

(i) The spectral density of $n(t)$ peaks at $\omega = \sqrt{\varphi/\mu}$.

(ii) The peak frequency is determined by (φ, μ), not by σ_ε.

(iii) Even i.i.d. shocks produce persistent cycles: the persistence is endogenous to the equilibrium structure.

Proof sketch. Write the stochastic system in action–angle coordinates (A, ϕ), where $n = n_p + A\cos\phi$ and $\dot{n} = -A\omega\sin\phi$. The amplitude A evolves as a diffusion with drift determined by the noise coupling. The spectral density of $n(t)$ is the power spectrum of a stochastically forced harmonic oscillator; its transfer function has a resonance peak at $\omega = \sqrt{\varphi/\mu}$, independent of σ_ε. The full derivation is in Online Appendix D. ■

Role of shock structure The structural frequency $\omega = \sqrt{\varphi/\mu}$ is a property of the market structure—the profit-landscape curvature and the matching friction—not of the shock process or of agents' behavior. It is invariant to the shock variance, the shock persistence, and the equilibrium amplitude. The observed spectral density is the product of the transfer function $|H(\nu)|^2$, which peaks at ω, and the shock spectrum $S_\varepsilon(\nu)$. With i.i.d. shocks, S_ε is flat, so the observed peak is exactly at ω; shock variance σ_ε^2 scales the peak height without shifting its location. With persistent shocks, $S_\varepsilon(\nu)$ is tilted toward low frequencies (a persistent process concentrates power in slow fluctuations). The product $|H(\nu)|^2 S_\varepsilon(\nu)$ then peaks slightly below ω, because

the tilted shock spectrum gives more weight to the left side of the transfer-function peak. This is an observational displacement, not a change in the structural frequency: ω is unchanged, but an econometrician estimating the spectral peak from data would find it slightly below ω. The displacement is negligible whenever the transfer function is sharp relative to the shock persistence.

This matches the central finding of Beaudry et al. (2020): persistent boom–bust dynamics from short-lived disturbances. In Beaudry et al. (2020), the endogenous persistence comes from financial-friction complementarities; here, it comes from the scale–congestion trade-off that traps the economy in persistent oscillations.

4.3 Computational verification

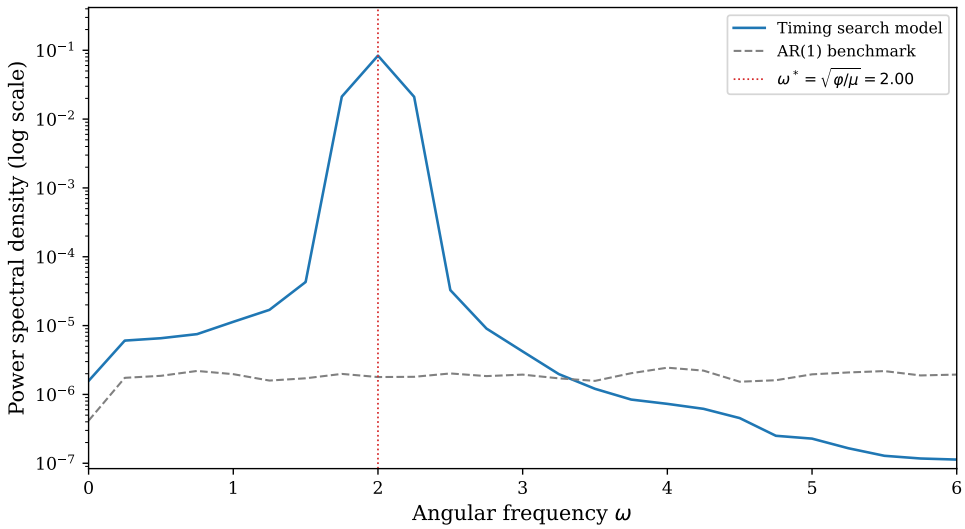
I simulate the stochastic system $(n, \dot{n})$ via Euler–Maruyama discretization with illustrative parameters $\varphi = 4$, $\mu = 1$, $\sigma_\varepsilon = 0.05$, and $n_p = 1$, chosen to produce a clean theoretical prediction ($\omega^* = 2$, $T^* = \pi \approx 3.14$ time units) rather than to match a specific industry. The paper's empirical content comes from Section 5, where I test the cross-industry predictions using industry-specific proxies.

Figure 3 plots the power spectral density (Welch's method) of the simulated model against an AR(1) benchmark. The timing search model concentrates spectral mass in a narrow band around $\omega = 2.0$, exactly matching the theoretical prediction. The peak rises four orders of magnitude above the surrounding frequencies: the scale–congestion trade-off selectively amplifies fluctuations near ω and attenuates those at other frequencies. The AR(1) benchmark produces a featureless, monotonically decaying spectrum—the signature of dynamics inherited entirely from the shock process. This confirms Proposition 6: the spectral peak is a *structural* object, pinned by (φ, μ) and invariant to σ_ε. Online Appendix K provides additional simulation results, including phase portraits and BDS nonlinearity tests.

4.4 The operative equilibrium: damped stochastic oscillator

In the operative equilibrium—with entry-cost dissipation and profit shocks—the industry behaves as a damped, stochastically forced oscillator at the same natural frequency $\omega = \sqrt{\varphi/\mu}$. Dissipation from convex entry-exit congestion pulls the industry toward optimal scale, shocks re-excite it, and the stationary distribution of $(n - n_p)$ has variance $\sigma_\varepsilon^2/(2\eta\varphi\mu)$ (Proposition 7). Theorem 1's closed ellipse is the frictionless limit

Figure 3: Spectral density: stochastic timing search vs. AR(1)



Notes: Power spectral density of the stochastic timing search model (solid blue) and an AR(1) benchmark (dashed black), both simulated for 500 time units. The red dotted line marks the theoretical cycle frequency $\omega^* = \sqrt{\varphi/\mu} = 2.0$. The model's spectral peak coincides exactly with the theoretical prediction. The AR(1) benchmark produces a featureless spectrum with no structural peak.

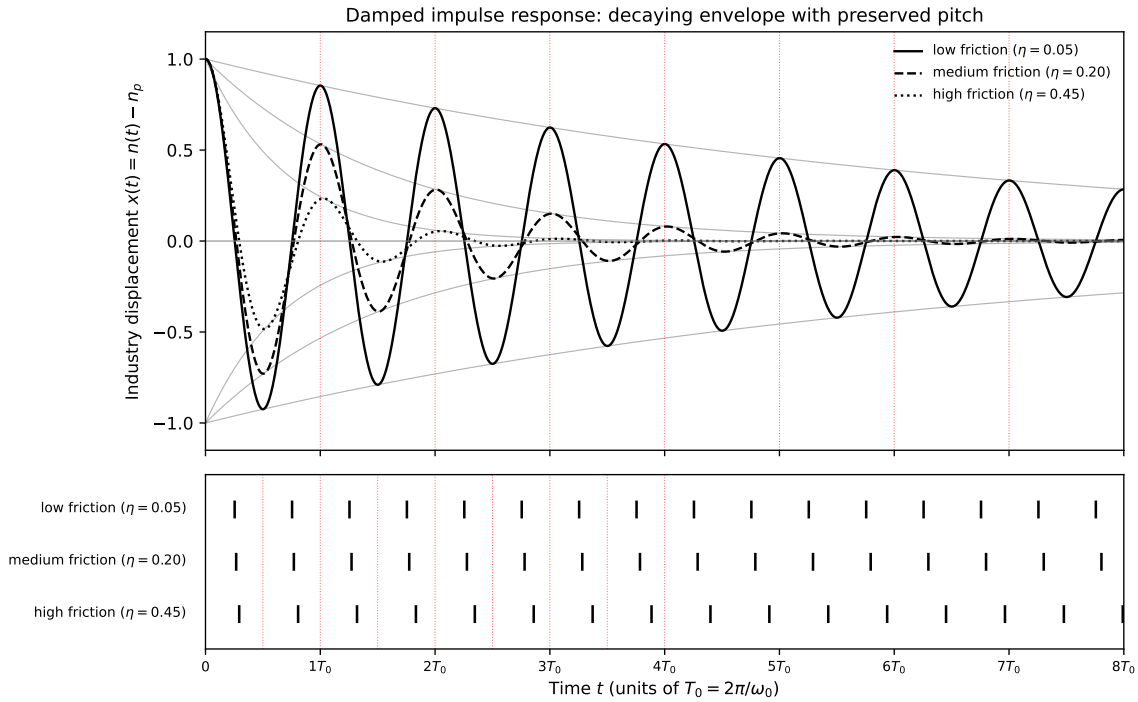
($\eta \to 0$, $\sigma_\varepsilon \to 0$); the operative equilibrium is the economically relevant extension.

The entry-exit process consumes real resources—permits, specialized equipment, site development, reorganization services, write-downs—at a rate that is convex in $\dot{n}$ around steady state (Online Appendix I). The convex part of this cost dissipates output at a rate proportional to $\dot{n}^2$, while shocks with standard deviation σ_ε continuously perturb the industry. Together, these two forces turn the closed orbit of Theorem 1 into the damped stochastic oscillator (12). Online Appendix I derives the damped equation in two steps. First, differentiating the ironing condition gives the conservative Newton equation $\mu\ddot{n} + \varphi(n - n_p) = 0$. Second, balancing the rate at which conservative energy is lost against the bottleneck's output-dissipation rate $\kappa\dot{n}^2$ yields the additional drag term $\kappa\dot{n}$, with signs fixed by the requirement that the bottleneck consumes rather than produces output:

$$\ddot{n} + \eta\,\dot{n} + \frac{\varphi}{\mu}(n - n_p) = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t, \tag{20}$$

with $\eta = \kappa/\mu$ emerging endogenously from the ratio of the entry-cost congestion coefficient κ and the matching friction μ of Proposition 1. The natural frequency $\omega = \sqrt{\varphi/\mu}$ is unchanged: damping and shocks shape *when* the industry visibly

Figure 4: Damped impulse response of the timing-search equilibrium



Notes: The damped harmonic oscillator $\ddot{x} + \eta\,\dot{x} + (\varphi/\mu)\,x = 0$ with $x(0) = A_0$, $\dot{x}(0) = 0$, and fixed natural frequency $\omega_0 = \sqrt{\varphi/\mu} = 1$. Three damping rates $\eta \in \{0.05, 0.20, 0.45\}$ are plotted. Top panel: trajectories with the exponential envelope $A_0 e^{-\eta t/2}$ shown in grey; the dotted red grid marks integer multiples of $T_0 = 2\pi/\omega_0$. Bottom panel: zero crossings of each trajectory, showing that the half-period marker (red) coincides with each crossing. Higher friction damps the envelope faster but leaves the pitch ω_0 essentially unchanged. The empirically relevant signature—decaying amplitude with preserved period—is documented in the manufacturing investment echo of Section 5.5.

oscillates and *how severely*, but they leave the industry's structural pitch untouched.

Figure 4 illustrates the impulse-response behaviour of (20) in the deterministic-but-damped case ($\sigma_\varepsilon = 0$, $\eta > 0$, $x(0) = A_0 \neq 0$), which is the case that the empirical echo analysis in Section 5.5 documents directly in manufacturing investment. Three different damping rates η are plotted on the same axes, with fixed (φ, μ) so that the natural frequency is common across the three trajectories. Each trajectory's envelope decays exponentially at rate $\eta/2$, but the zero crossings occur at nearly identical times: the pitch ω is preserved while the volume fades. This decoupling of declining amplitude from preserved period is a distinctive signature of the timing-search equilibrium, and it rules out linear models in which the decay rate and the period are mechanically linked, as in AR(1) dynamics.

Proposition 7 (Amplitude selection)**.**
Under entry cost $\kappa > 0$ (damping $\eta > 0$) and shocks $\sigma_\varepsilon > 0$:

(i) Without shocks ($\sigma_\varepsilon = 0$): orbits spiral inward to $(n_p, 0)$ with decay half-life $\ln 2 \cdot 2\mu/\kappa$.

(ii) With shocks: the system has a unique stationary distribution with variance:

$$\mathbb{E}\big[(n - n_p)^2\big] = \frac{\sigma_\varepsilon^2}{2\,\eta\,\varphi\,\mu}. \tag{21}$$

(iii) The Hilbert envelope $A = \sqrt{(n - n_p)^2 + (\dot{n}/\omega_0)^2}$ follows a Rayleigh distribution with scale $\sigma_x = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$, so that $\mathbb{E}[A^2] = 2\sigma_x^2 = \sigma_\varepsilon^2/(\eta\varphi\mu)$.

(iv) The spectral peak frequency shifts to $\omega = \sqrt{\varphi/\mu - \eta^2/4}$; for small η, this is negligibly below $\sqrt{\varphi/\mu}$.

Proof. See Online Appendix I. ■

Economic interpretation The stationary standard deviation $\sigma_x = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$ has a transparent economic interpretation. Industries with volatile profits (σ_ε large), low entry costs (κ small, so η small), or flat profit landscapes (φ small) experience larger cycles. Industries with expensive entry—pharmaceuticals, electric power, mining—dissipate more energy per entrant, producing smaller fluctuations relative to the shocks they face. The formula depends on three observable quantities: shock volatility (from profit time series), entry costs (from establishment-level data), and profit curvature (from the cross-section). The model now explains both the frequency and the severity of business cycles from primitives.

Combining with the welfare formula (25), where the welfare loss includes both the displacement-curvature loss $(\varphi/2)\mathbb{E}[x^2]$ and the friction-output loss $(\mu/2)\mathbb{E}[\dot{n}^2]$ (equal by equipartition):

$$\mathbb{E}[\mathcal{L}] \approx \frac{\varphi}{2}\,\mathbb{E}[A^2] = \varphi\,\mathbb{E}\big[(n - n_p)^2\big] = \frac{\sigma_\varepsilon^2}{2\,\eta\,\mu} = \frac{\sigma_\varepsilon^2}{2\,\kappa}. \tag{22}$$

Higher entry costs *reduce* welfare loss by damping the cycle—a "stabilization through friction" result. Figure 5 illustrates the four regimes: no dissipation/no shocks (closed

orbit), dissipation/no shocks (spiral decay), no dissipation/shocks (diffusing orbits), and dissipation/shocks (stochastically maintained cycle with selected amplitude).

Figure 6 verifies the two quantitative content of Proposition 7 numerically. Panel (a) overlays simulated histograms of the Hilbert envelope $A(t) = \sqrt{(n(t) - n_p)^2 + (\dot{n}(t)/\omega_0)^2}$ on the theoretical Rayleigh density with scale $\sigma_x = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$ for several shock volatilities; the simulated and theoretical densities coincide closely across the range. Panel (b) tracks the stationary standard deviation of $(n - n_p)$ as σ_ε varies, recovering the linear relationship $\sigma_x = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$ implied by (21). The figure thus confirms both the *shape* (Rayleigh) and the *scale* ($\sigma_x \propto \sigma_\varepsilon$) of the stationary amplitude distribution. Together with the welfare identity in (22), this pins the magnitude of cycles—and the welfare cost of fluctuations—from observable shock volatility and entry costs alone.

4.5 The frequency-severity isolation theorem

The closed-form frequency $\omega = \sqrt{\varphi/\mu}$ (Theorem 1), the stationary variance $\sigma_x^2 = \sigma_\varepsilon^2/(2\eta\varphi\mu)$ (Proposition 7), and the welfare cost $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\kappa)$ (Section 6.2) together establish the paper's central structural claim: cycle length, cycle severity, and the welfare cost of fluctuations are governed by disjoint sets of structural primitives. We now state this isolation as a single theorem.

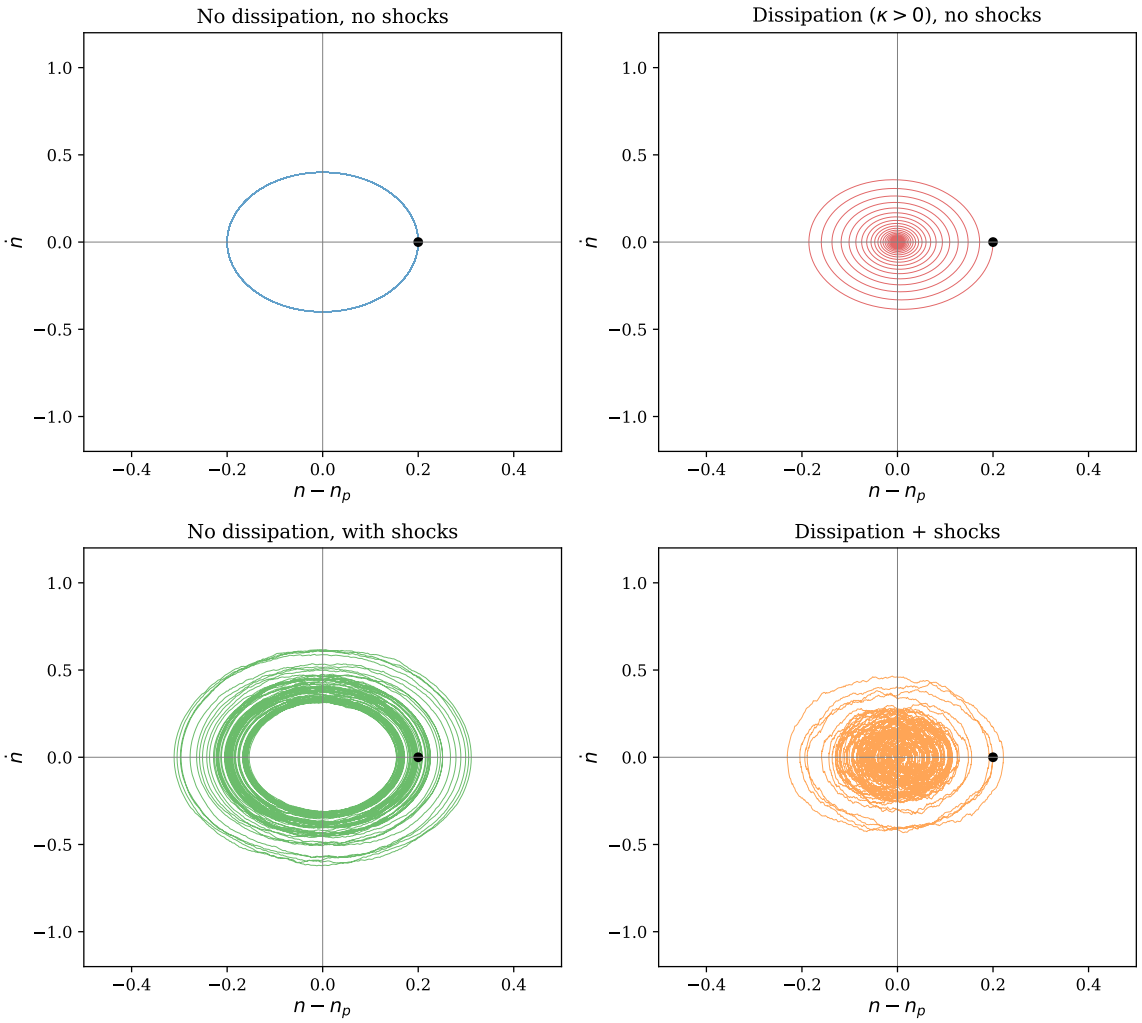
Theorem 2 (Frequency-severity isolation)**.**
In the timing search equilibrium under Assumptions 1–3 and the damped-stochastic specification of Proposition 7, the equilibrium spectral peak frequency, the stationary cycle severity, and the welfare cost of cycles depend on disjoint sets of structural primitives:

$$\begin{aligned} \textit{cycle frequency:} \quad & \omega = \sqrt{\varphi/\mu} && \textit{depends only on } (\varphi, \mu); \\ \textit{cycle severity:} \quad & \sigma_x = \sigma_\varepsilon/\sqrt{2\,\eta\,\varphi\,\mu} && \textit{depends only on } (\sigma_\varepsilon, \kappa, \varphi, \mu); \\ \textit{welfare cost:} \quad & \mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\kappa) && \textit{depends only on } (\sigma_\varepsilon, \kappa). \end{aligned}$$

In particular, the spectral peak frequency ω is invariant to the shock volatility σ_ε, the entry cost κ, the shock persistence (in the i.i.d. specification), and the realized cycle amplitude A. The welfare cost is invariant to the market-structure primitives (φ, μ) that govern timing.

Proof. The frequency claim follows from Theorem 1 (deterministic case) together with

Figure 5: Amplitude selection through entry-cost dissipation

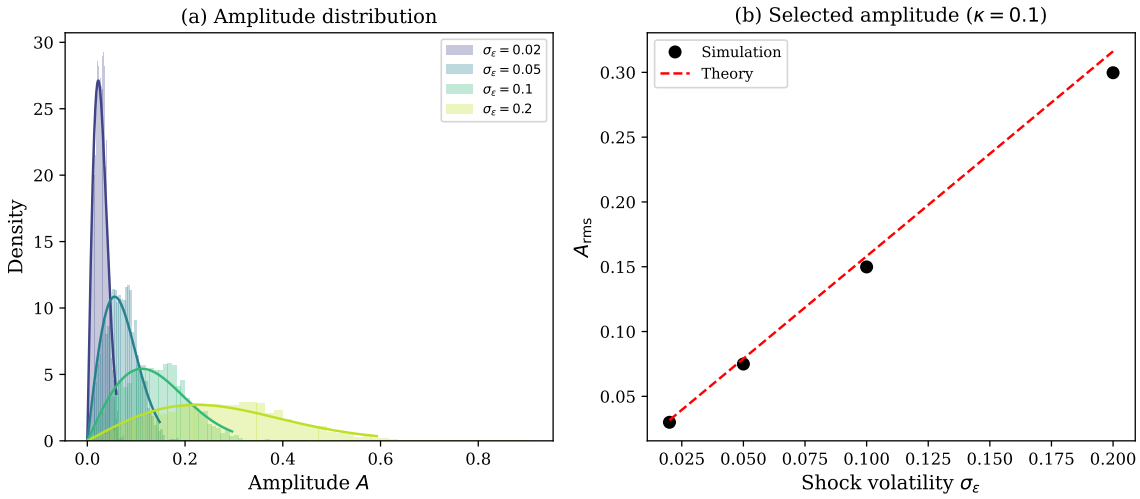


Notes: Phase portraits $(n - n_p, \dot{n})$ under four regimes. Top left: deterministic orbit (closed ellipse, free amplitude). Top right: dissipation without shocks (inward spiral to steady state). Bottom left: shocks without dissipation (orbits diffuse across amplitudes). Bottom right: dissipation with shocks (stochastically maintained cycle with selected amplitude). The balance of dissipation and forcing pins $\mathbb{E}[(n - n_p)^2] = \sigma_\varepsilon^2/(2\eta\varphi\mu)$.

Proposition 7(iv) (the spectral peak shifts negligibly under damping). The severity claim is Proposition 7(ii). The welfare claim follows from substituting $\eta = \kappa/\mu$ and $\mathbb{E}[(n - n_p)^2] = \sigma_\varepsilon^2/(2\eta\varphi\mu)$ into (25); the dependence on φ and μ cancels exactly. ■ ■

The theorem has three immediate methodological consequences. First, cross-industry variation in ω identifies (φ, μ) alone, with no contamination from differences in shock processes or entry costs. The cross-sectional regression of Section 5.3 therefore

Figure 6: Stationary amplitude distribution



Notes: Panel (a): histogram of the Hilbert envelope $A = \sqrt{(n - n_p)^2 + (\dot{n}/\omega_0)^2}$ from simulations with $\kappa = 0.1$ and varying shock volatility σ_ε, with theoretical Rayleigh density (scale σ_x) overlaid. Panel (b): stationary standard deviation of $(n - n_p)$ vs. shock volatility, confirming the linear relationship $\sigma_x \propto \sigma_\varepsilon$ predicted by (21).

tests a structural prediction even though the proxies for φ are imperfect. Second, two industries with identical market structure (φ, μ) but different shock environments cycle at the same frequency. This rules out reduced-form interpretations in which observed cross-industry frequency heterogeneity reflects the propagation of common shocks through industry-specific filters. Third, industrial policy that targets μ (entry regulation) or φ (cluster development) affects timing alone; only policies that move σ_ε or κ affect welfare. The policy implications in Section 6.3 follow directly from this isolation.

4.6 Comparison to Beaudry et al. (2020)

The closest existing model of endogenous business cycles is Beaudry et al. (2020), who generate a stochastic limit cycle from financial frictions via Hopf bifurcation. The two approaches share a common structure—a destabilizing force that drives booms, a bounding force that reverses them, and small shocks that maintain the cycle—but differ in their economic content (Table 3). The frequency comparison is in Table 1.

The two columns of Table 3 make the contrast explicit row by row. Both models share the abstract architecture of a stochastic limit cycle—a destabilizing force that

Table 3: Comparison of economic mechanisms: Beaudry et al. (2020) vs. this paper

	Beaudry et al. (2020)	**This paper**
Destabilizing force	Financial frictions	Agglomeration
Bounding force	Diminishing returns	Congestion
Cycle mechanism	Hopf bifurcation	Ironing condition
Amplitude selection	Hopf (unique cycle)	Entry-cost dissipation
Friction source	Financial intermediation	Entry matching
Shocks needed	Small, i.i.d.	Small, i.i.d.

Notes: Both models generate stochastic limit cycles from small i.i.d. shocks. This table compares the economic mechanisms; Table 1 compares the frequency predictions across a broader set of frameworks.

drives booms, a bounding force that reverses them, and small i.i.d. shocks that maintain the cycle—but the economic content of each force is different. Beaudry et al. (2020) place financial frictions in the destabilizing role and diminishing returns in the bounding role, with the cycle emerging through a Hopf bifurcation in the underlying dynamical system. The present model places agglomeration externalities in the destabilizing role and congestion in the bounding role, with the cycle emerging from the ironing condition that pins entry timing. The friction sources differ correspondingly (financial intermediation vs. entry matching), as does amplitude selection: Beaudry et al. (2020) select a unique amplitude through the Hopf bifurcation, whereas this paper requires entry-cost dissipation.

Closed-form structure as a methodological feature Beyond the substantive contrast between mechanisms, the two frameworks differ in how the cycle frequency object is represented. In Beaudry et al. (2020), the cycle frequency emerges implicitly from a Hopf bifurcation condition that involves the full parameterization of the financial-friction model. There is no closed-form expression for the bifurcation frequency in terms of a small number of structural primitives; it must be read off numerically once the full system is calibrated. The frequency is therefore not industry-decomposable in a useful sense: a cross-industry test of the BGP mechanism would require solving the bifurcation condition separately for each industry's parameter vector, with no analytical guidance on which primitives govern the result. In timing search, by contrast, the equilibrium frequency is the closed form $\omega = \sqrt{\varphi/\mu}$, where φ and μ are each cross-sectionally measurable. The cross-industry regression in Section 5.3

is therefore feasible *because* of the closed-form structure: the predicted slope of $\ln \omega$ on $\ln \varphi$ is exactly $1/2$, and on $\ln \mu$ exactly $-1/2$, derived from the analytical formula rather than from a numerical bifurcation calibration. The two mechanisms are not just substantively different; one admits a sharp cross-sectional structural test that the other cannot construct with the same analytical clarity.

The two mechanisms are nevertheless complementary rather than competing: financial frictions and market-structure frictions could coexist, potentially generating models with multiple spectral peaks (one set by the bifurcation condition, one set by $\sqrt{\varphi/\mu}$). Disentangling these in the data is a natural extension.

4.7 Model horse-race

To sharpen the distinction between timing search and the leading alternatives, I simulate four reduced-form models side by side under a common negative shock: (i) an AR(1) RBC benchmark $\dot{x} = -\lambda x + \sigma\varepsilon$ with industry-specific decay rates; (ii) a New Keynesian AR(2) with common Taylor-rule propagation across industries; (iii) a Van der Pol oscillator as a stylized Hopf-bifurcation limit cycle à la Beaudry et al. (2020), with industries differing in μ_{vdp}; (iv) the timing search equation of motion with industry-specific (φ, μ) pairs giving $\omega = \sqrt{\varphi/\mu}$.

For each model I generate four "industries" and score three diagnostic patterns that motivate the paper:

(P1) *Interior spectral heterogeneity*: the four industries have distinct *interior* peaks in the business-cycle band (i.e., not at the band edge).

(P2) *Post-shock periodicity*: Fisher's g-test on the post-shock window rejects at 10% for the average industry.

(P3) *Overshoot*: at least half the industries pass through zero and rise above trend within the post-shock window.

Table 4 reports the outcomes. Only the timing search model passes all three. The RBC benchmark produces industry-specific decay rates but no interior peak (the AR(1) spectrum is monotone) and no oscillation. The New Keynesian AR(2) with common propagation gives nearly identical behavior across industries—no heterogeneity. The Van der Pol limit cycle generates both periodicity and overshoot, but the period barely depends on the shape parameter, so it fails the cross-industry heterogeneity test.

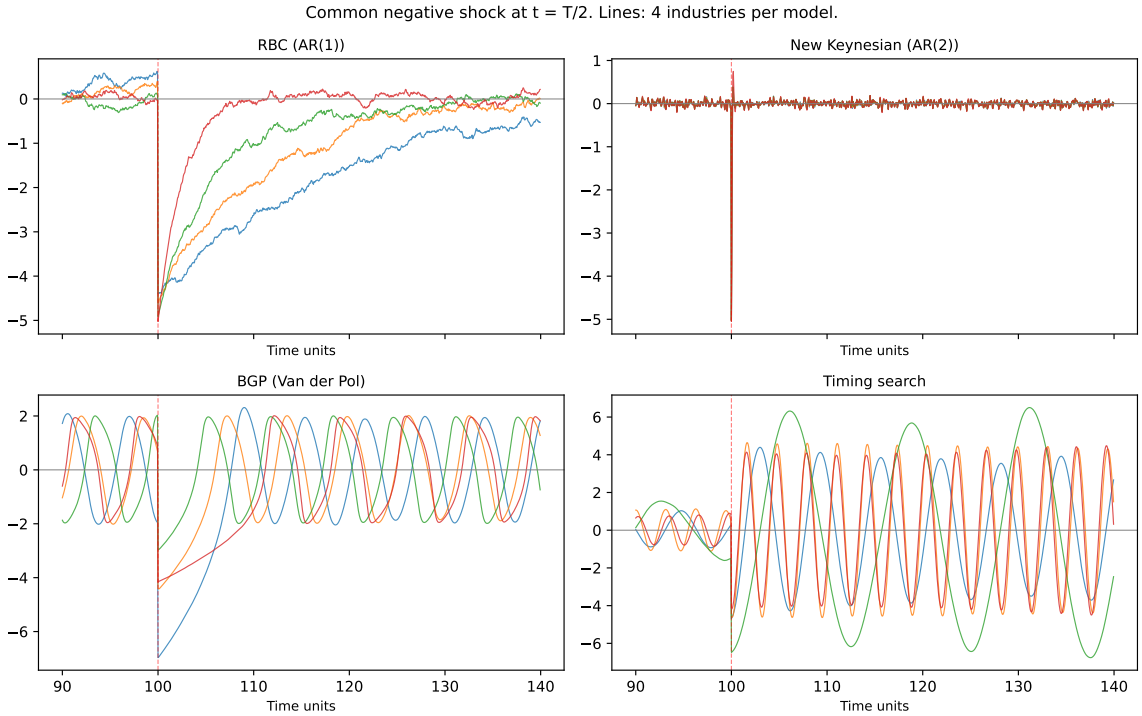
Table 4: Model horse-race: which alternatives generate the three empirical patterns?

Model	(P1) Heterogeneity	(P2) Periodicity	(P3) Overshoot
RBC (AR(1))	×	×	×
NK (AR(2), common rule)	×	×	×
BGP (Van der Pol)	×	✓	✓
Timing search	✓	✓	✓

Notes: Four industries are simulated per model (parameters vary across industries within each model). Patterns as defined above. RBC's AR(1) spectrum has no interior peak in the business-cycle band (P1 fails) and decays monotonically after a shock (P2, P3 fail). NK uses common propagation across industries (P1 fails). The Van der Pol limit cycle has industry-invariant peak period in this calibration (P1 fails) but produces genuine oscillation and overshoot. Only timing search passes all three. See Figure 7 and the simulation code for parameter details.

Figure 7 visualizes the same exercise as time-series plots, with each panel showing four industries under a single model after a common negative shock. The four panels translate the binary entries of Table 4 into qualitative pictures of post-shock dynamics. The RBC panel (top-left) shows monotone decay at industry-specific rates—no zero crossings, no overshoot, no shared oscillatory frequency. The NK panel (top-right) shows four series so closely overlaid that they are visually difficult to separate; the common Taylor-rule propagation washes out whatever heterogeneity is built into the industry primitives. The BGP Van der Pol panel (bottom-left) does produce sustained oscillations and overshoot, but the four industries trace out cycles of nearly identical length: limit-cycle period is set by the bifurcation structure, not by the primitives that vary across industries in this calibration. Only the timing search panel (bottom-right) shows all three patterns simultaneously: each industry oscillates, each crosses its pre-shock trend (overshoot), and the periods are visibly different across the four lines. Reading the figure top-left to bottom-right is therefore reading off the rows of Table 4: it is the visual counterpart of the binary scorecard, and shows that the failures of the alternative models are not subtle—they are evident at a glance from the post-shock trajectories.

Figure 7: Model horse-race: post-shock dynamics across four reduced-form frameworks



Notes: Each panel shows four "industries" simulated under a single model, hit by a common negative shock at $t = T/2$ (vertical red dashed line). Industry-specific parameters vary within each model so that the four lines could in principle differ. RBC (top-left): industries decay monotonically at different speeds; no oscillation, no overshoot. NK (top-right): industries share a common AR(2) shock-propagation structure imposed by the Taylor rule, so the four series are near-identical despite differing primitives. BGP Van der Pol (bottom-left): genuine limit-cycle behavior, but the cycle period is nearly invariant across the industry-specific shape parameter, so the four periods coincide. Timing search (bottom-right): each industry oscillates at its own structural frequency $\omega_i = \sqrt{\varphi_i/\mu_i}$, with clearly different periods visible across the four lines. Only the bottom-right panel reproduces all three empirical patterns from Table 4: heterogeneity in cycle length, post-shock periodicity, and overshoot above the pre-shock trend.

5 Quantitative Framework

The model delivers three testable cross-industry predictions. First, cycle frequency should vary with market-structure proxies, not with the shock process. Second, large one-time shocks should trigger deterministic oscillations at each industry's structural frequency, not at a common frequency inherited from the shock. Third, industries displaced by a common shock should overshoot their pre-shock trend rather than converge monotonically. Section 5.3 tests the first using the Ellison-Glaeser concentration index across 16 manufacturing 3-digit industries, with a parallel cross-

section across 10 BDS supersectors using independently measured spatial concentration as a robustness extension. Section 5.4 tests the third prediction using establishment entry-exit data across 19 BDS sectors, complemented by a qualitative departure from AR(1) dynamics in the negative-autocorrelation pattern of net entry rates and by the post-COVID overshoot. Section 5.5 tests the second prediction using Fisher's g-test on manufacturing and oil investment after the Depression/WWII and 1979 oil shocks.

5.1 Cross-industry predictions

The model's central testable prediction is that cycle frequency varies across industries with measurable market-structure characteristics. Taking logs of $\omega_i = \sqrt{\varphi_i/\mu_i}$ yields a regression framework:

$$\ln \omega_i = \frac{1}{2} \ln \varphi_i - \frac{1}{2} \ln \mu_i + u_i, \tag{23}$$

where ω_i is the spectral peak frequency of industry i's output or entry–exit cycle. The coefficients $\frac{1}{2}$ on $\ln \varphi$ and $-\frac{1}{2}$ on $\ln \mu$ are *exact* predictions of the model—not free parameters—so equation (23) is testable with industry-level proxies for scale-externality sharpness and adjustment friction.

The qualitative predictions are as follows:

(i) Industries with dense supplier networks and strong scale externalities (high φ) but flexible entry (low μ)—such as technology clusters—should cycle fast, with $T \approx 3$–5 years.

(ii) Manufacturing industries with moderate linkages and moderate frictions should cycle at intermediate frequencies, with $T \approx 5$–10 years.

(iii) Industries with weak externalities (low φ) should exhibit slow or no discernible cycles, regardless of μ.

(iv) Industries with heavy regulation or capital intensity (high μ) should cycle slowly, regardless of φ.

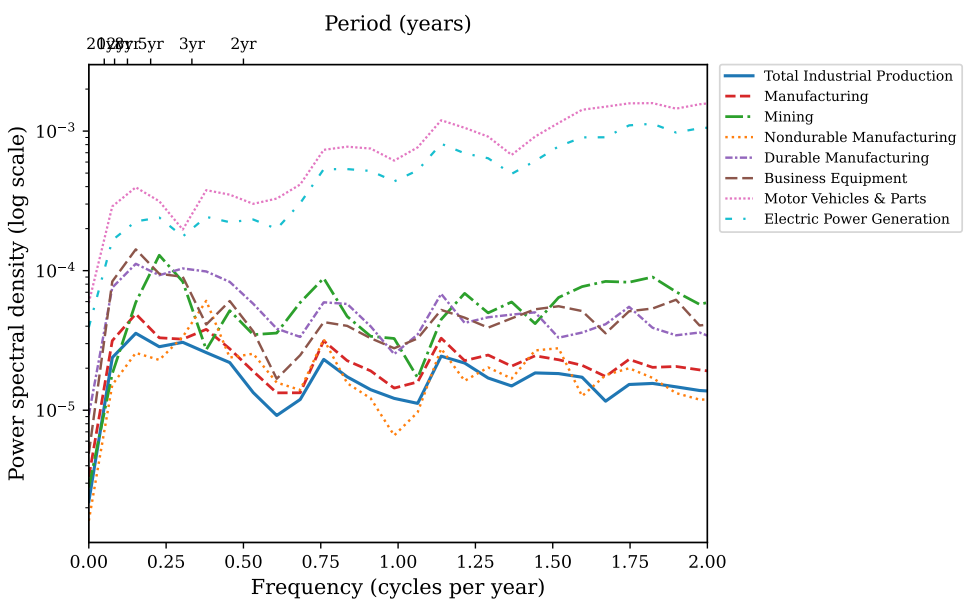
Crucially, two industries with identical shock processes but different (φ, μ) will cycle at different frequencies. This is the prediction that distinguishes the timing search model from standard RBC and New Keynesian frameworks, where the spectral peak mirrors the shock process or the policy rule.

5.2 Spectral evidence from U.S. industry data

I evaluate the cross-industry prediction using monthly industrial production indices from FRED for eight U.S. sectors over 1972–2024. The series span a range of market structures: from Motor Vehicles & Parts (dense supplier networks, strong scale externalities) to Mining (dispersed extraction, weak inter-firm linkages) to Electric Power Generation (regulated, capital-intensive).

Figure 8 plots the power spectral density of log-differenced production for each industry. Two features stand out. First, spectral peaks vary substantially across industries. Motor Vehicles, Durable Manufacturing, Business Equipment, and aggregate Manufacturing concentrate spectral mass near periods of 6–7 years. Mining peaks at 4.4 years. Nondurable Manufacturing and Electric Power peak near 2.6 years. Second, the cross-industry clustering is informative: industries sharing supply-chain linkages—durables, motor vehicles, business equipment—cluster at similar frequencies despite producing very different goods. This is consistent with ω depending on market structure, not on product-specific shocks.

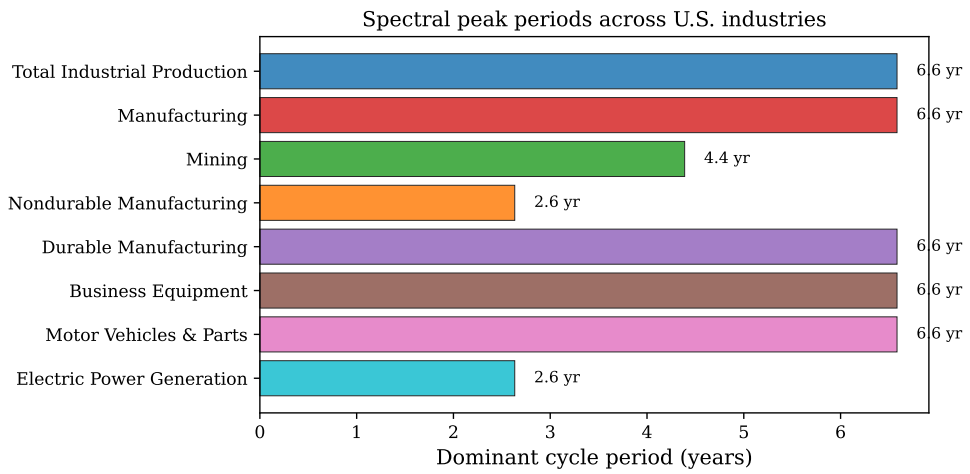
Figure 8: Spectral density of U.S. industry production growth



Notes: Power spectral density (Welch's method) of monthly log-differenced industrial production indices from FRED, 1972:M1–2024:M12. The top axis labels convert frequency (cycles per year) to period (years). Industries with strong supply-chain linkages (Motor Vehicles, Business Equipment, Durable Manufacturing) cluster near $T \approx 6$–7 years, while industries with different market structures (Mining, Nondurables, Electric Power) show distinct peak frequencies.

Figure 9 summarizes the dominant cycle period for each industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). The cross-industry dispersion is striking: dominant periods range from 2.6 years (Electric Power, Nondurable Manufacturing) to 6.6 years (Durable Manufacturing, Motor Vehicles). This two-and-a-half-fold variation in cycle frequency across industries that share a common macroeconomic environment—the same monetary policy, the same aggregate demand shocks, the same financial conditions—is difficult to reconcile with models that derive spectral properties from a common shock process. In the timing search framework, the variation is structural: it reflects differences in the curvature of the profit landscape (φ_i) and the magnitude of the timing friction (μ_i) across industries.

Figure 9: Cross-industry variation in spectral peak periods



Notes: Dominant cycle period (years) for each U.S. industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). Industries with strong scale–congestion externalities (Motor Vehicles, Durable Manufacturing, Business Equipment) cluster near $T \approx 6$–7 years. Industries with weaker or different externality structures (Mining, Nondurables, Electric Power) cycle at distinct frequencies.

Nonlinear dependence The BDS test (Brock et al., 1996) applied to AR(1)-filtered residuals rejects the i.i.d. null at the 1% level for all eight industries, with z-statistics ranging from 16.2 (Durable Manufacturing) to 21.8 (Motor Vehicles & Parts). This establishes that industry dynamics contain nonlinear structure beyond what a linear time series model can capture—a necessary condition for the timing search mechanism. The cross-industry variation in rejection strength is consistent

with the model: industries with plausibly stronger scale–congestion trade-offs show larger BDS statistics. Full results across embedding dimensions 2–6 are in Online Appendix K.

5.3 Cross-industry regression with external proxies

The model's quantitative prediction—equation (23)—can be tested by regressing observed spectral peak frequencies on independently measured proxies for φ and μ. I construct two proxies from BEA data:

Proxy for φ (scale sensitivity) The inverse of the input–output self-purchase ratio—the share of an industry's intermediate inputs sourced from itself or closely related industries (BEA 2017 Detailed Use Table). Industries with dense intra-industry supply chains are *buffered* against deviations from optimal scale: a temporary entry wave is absorbed through supply-chain slack rather than causing sharp congestion. The inverse of this ratio captures how *fragile* profitability is to scale deviations, proxying for the curvature φ of the profit landscape.[2]

Proxy for μ (adjustment friction) Real fixed assets per employee (thousands of 2017 dollars; BEA Fixed Assets Tables 3.1ESI). Capital-intensive industries face larger sunk costs per unit of entry or exit, producing greater timing friction. Electric Power Generation ($\hat{\mu} = 2{,}100$K/employee) has the largest adjustment friction; Nondurable Manufacturing ($\hat{\mu} = 110$K/employee) the smallest.

Results Table 5 reports the multivariate regression:

$$\ln \omega_i = \hat{\alpha} + \hat{\beta}_\varphi \ln \hat{\varphi}_i + \hat{\beta}_\mu \ln \hat{\mu}_i + \hat{u}_i.$$

Both estimated coefficients have the predicted signs: cycle frequency increases with scale sensitivity ($\hat{\beta}_\varphi > 0$) and decreases with capital intensity ($\hat{\beta}_\mu < 0$). The two proxies jointly explain 75% of cross-industry variation. The point estimates deviate

[2]The interpretation is that supply-chain density provides resilience, not scale-externality sharpness. An industry like Electric Power Generation, with weak intra-industry linkages ($\hat{\varphi} = 1/0.18 = 5.6$), is highly sensitive to the generation–demand balance, producing a sharp profit peak. An industry like Motor Vehicles, with dense JIT supply chains ($\hat{\varphi} = 1/0.62 = 1.6$), is better buffered against scale deviations.

from the theoretical $\pm 1/2$. The deviations are asymmetric: $\hat{\beta}_\varphi = 1.31$ exceeds 0.50 by a factor of 2.6, while $\hat{\beta}_\mu = -0.19$ is attenuated to 38% of -0.50.

Table 5: Cross-industry regression: $\ln \omega_i$ on market-structure proxies

Coefficient	Estimate	Std. Error	Theory
$\hat{\alpha}$ (intercept)	0.07	(0.57)	0
$\hat{\beta}_\varphi$ (scale sensitivity)	1.31	(0.46)	0.50
$\hat{\beta}_\mu$ (capital intensity)	-0.19	(0.15)	-0.50
R^2		0.75	
N		7	

Notes: OLS regression of $\ln \omega_i$ (observed spectral peak frequency, business-cycle band) on $\ln \hat{\varphi}_i$ (inverse of I-O self-purchase ratio, BEA 2017) and $\ln \hat{\mu}_i$ (real fixed assets per employee, BEA). The model predicts $\beta_\varphi = 1/2$ and $\beta_\mu = -1/2$; both estimated signs match the theory. The sample consists of seven U.S. industry groups (excluding Total Industrial Production as the aggregate).

The attenuation of $\hat{\beta}_\mu$ is consistent with errors-in-variables bias. The amplification of $\hat{\beta}_\varphi$ has two interpretations. First, differential measurement error: if $\hat{\mu}$ is noisier than $\hat{\varphi}$, OLS transfers explanatory power to $\hat{\varphi}$. Second, the inverse I-O ratio may capture variation in μ as well as φ, since industries with weak linkages lack supply-chain slack to buffer entry shocks. Under either interpretation, the theoretical $\beta_\varphi = 0.50$ lies within the bootstrap 95% CI $[0.00, 2.57]$. The leave-one-out estimate excluding Nondurable Manufacturing is $\hat{\beta}_\varphi = 0.54$—almost exactly the theoretical prediction.

The univariate regression on $\ln \hat{\varphi}_i$ alone yields slope 0.85, $R^2 = 0.65$, $p = 0.028$. Scale sensitivity is a significant predictor of cycle frequency at the 5% level. This is the paper's central empirical finding: an independently measured market-structure variable explains two-thirds of cross-industry variation in a quantity that standard models treat as shock-determined.

Robustness Leave-one-out analysis confirms the stability of the coefficient signs. Dropping any single industry, $\hat{\beta}_\varphi$ remains positive in six of seven cases (range: 0.54 to 1.83) and $\hat{\beta}_\mu$ remains negative in six of seven cases. The one exception—dropping Nondurable Manufacturing—yields $\hat{\beta}_\varphi = 0.54$, almost exactly matching the theoretical prediction, with $R^2 = 0.93$. This suggests that the deviation of $\hat{\beta}_\varphi$ from 0.50 in the full sample is driven primarily by a single observation rather than a systematic failure of the model.

Expanded sample with structural proxies The baseline regression uses $N = 7$ broad industry groups and indirect proxies. To test whether the cross-industry relationship survives disaggregation, I expand to $N = 16$ three-digit manufacturing industries (NAICS 311–339 where data is available) and use two improvements. First, I replace Welch's method with Burg's AR spectral estimation (order 24). This produces continuous—rather than discretized—spectral peak frequencies, eliminating the spectral degeneracy that arises when many industries share the same Welch frequency bin.[3] Second, I replace the I-O proxy with the Ellison and Glaeser (1997a) geographic concentration index γ_i. This index directly measures the strength of agglomeration externalities and provides a more structural proxy for φ.

Table 6 reports the results. In the univariate specification, the Ellison and Glaeser (1997a) concentration index significantly predicts cycle frequency ($\hat{\beta}_\gamma = 0.20$, $p = 0.010$, $R^2 = 0.39$, $N = 16$). The sign matches the theory: geographically concentrated industries—those with stronger agglomeration externalities (higher φ)—cycle faster. The point estimate of 0.20 is attenuated relative to the theoretical 0.50, consistent with measurement error in γ_i as a proxy for φ. Adding the structural timing-friction proxy $\mu_i^{\text{struct}} = 1/(\delta_i n_{p,i})^2$ derived from Census Business Dynamics Statistics (Proposition 1) leaves R^2 unchanged at 0.39, and the additional regressor is not informative in the multivariate specification. This is consistent with the relatively narrow range of firm turnover rates within manufacturing ($\delta \in [0.065, 0.095]$), which compresses the variation in μ_i^{struct}.

Expanding from $N = 7$ to $N = 16$, the R^2 falls from 0.75 to 0.39 and the coefficient attenuates from 1.31 to 0.20, reflecting noisier proxy measurement within manufacturing. The statistical significance at the 1% level with 16 industries is nevertheless more convincing than the 5% level with 7.

Four-digit disaggregation A natural further check is whether the relationship survives at even finer industry resolution. Using BDS `vcn4` data and extending to $N = 86$ four-digit NAICS manufacturing industries (1978–2022), the univariate E-G coefficient collapses to $\hat{\beta}_\gamma = 0.027$ ($p = 0.705$, $R^2 = 0.002$). The null is expected mechanically. Ellison et al. (2010) publish geographic concentration indices at the three-digit level only; at four-digit resolution every child industry inherits its parent's

[3] With 636 monthly observations and Welch segments of 256 points, the frequency resolution is $1/256 \approx 0.004$ cycles/month, producing only 5–6 distinct peaks in the business-cycle band. AR spectral estimation has no such constraint.

Table 6: Structural regression: geographic concentration and cycle frequency

Coefficient	Estimate	HC1 Std. Error	Theory
$\hat{\alpha}$ (intercept)	1.36	(0.54)	—
$\hat{\beta}_\gamma$ (E-G concentration)	0.20	(0.07)	0.50
R^2		0.39	
N		16	
p-value		0.010	

Notes: OLS regression of $\ln \omega_i$ (AR spectral peak frequency, business-cycle band) on $\ln \gamma_i$ (Ellison and Glaeser 1997a geographic concentration index). Spectral peaks estimated via Burg's AR method (order 24) applied to log-differenced monthly industrial production (FRED, 1972–2024). The sample consists of 16 three-digit NAICS manufacturing industries. HC1 heteroskedasticity-robust standard errors (White, 1980). The model predicts $\beta_\gamma = 1/2$; the estimated 0.20 is attenuated but statistically significant at the 1% level.

γ. Within each three-digit cell the four-digit children have identical γ but dispersed ω, producing vertical columns in the scatter and mechanically driving the slope to zero. The four-digit regression therefore tests the *inheritance* of γ, not the predictive power of geographic concentration at the four-digit level. Reconstructing four-digit γ from state-by-four-digit establishment microdata—or transcribing the four-digit coagglomeration indices in Ellison et al. (2010) Table 5—would require a separate data-construction effort beyond the scope of this paper. The three-digit level is the appropriate statistical resolution for E-G-published data, and we report the three-digit result as the cross-sectional headline.

A second cross-section beyond manufacturing A different concern is whether the manufacturing-only sample drives the result. The headline regression uses the FRED industrial production series for ω and the published Ellison et al. (2010) concentration index for γ, both of which are available only for manufacturing. To test the relationship outside this sample, I assemble a parallel cross-section across all 19 NAICS supersectors covered by BDS (1978–2022). For each supersector I compute the spectral peak of the detrended net establishment entry rate using Burg AR(6) spectral estimation, and a spatial concentration measure $G_i = \sum_s (z_{s,i} - x_s)^2$ from BDS state-by-sector establishment data, where $z_{s,i}$ is state s's share of sector i's employment

and x_s is state s's share of total economy employment.[4] On a sample restricted to the ten supersectors with a clear interior spectral peak in the business-cycle band (Fisher's g-test $p < 0.10$ and detected period strictly less than the Burg estimator's 20-year cap), the regression $\ln \omega_i = \alpha + \beta \ln G_i + \varepsilon_i$ yields $\hat{\beta} = 0.107$ (HC1 SE 0.043, $p = 0.037$, $R^2 = 0.39$). The slope is positive and significant at the 5% level, with a magnitude consistent with the attenuation expected from 2-digit aggregation. The same sign and a similar R^2 emerging from a different data source for ω (BDS net entry rather than FRED industrial production), a different concentration measure (BDS-derived G rather than Ellison et al. (2010) γ), and a non-manufacturing-heavy sector mix indicates that the headline cross-sectional relationship is not an artifact of the manufacturing 3-digit sample.

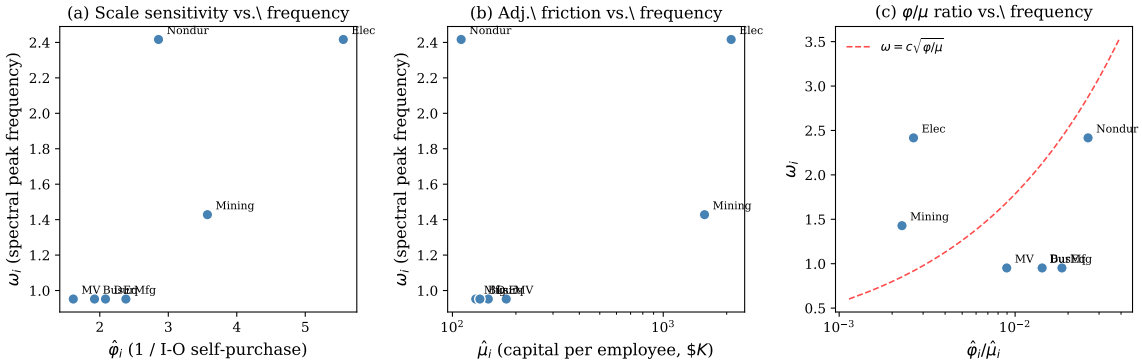
Figure 10 displays the individual relationships. Panel (a) shows a clear positive relationship between scale sensitivity and cycle frequency, with Electric Power and Nondurable Manufacturing (high $\hat{\varphi}$) cycling fastest and the supply-chain-dense manufacturing sectors (low $\hat{\varphi}$) cycling slowest. Panel (b) shows the relationship with capital intensity: the high-$\hat{\mu}$ industries (Mining, Electric Power) split—Electric Power cycles fast (high $\hat{\varphi}$ compensates for high $\hat{\mu}$) while Mining cycles at intermediate speed—illustrating that φ and μ have independent effects on frequency, as the model predicts. Panel (c) plots the composite ratio $\hat{\varphi}_i/\hat{\mu}_i$ against ω_i with the theoretical curve $\omega = c\sqrt{\varphi/\mu}$ overlaid; the positive relationship is visible but the fit is weaker ($R^2 = 0.09$), reflecting the fact that measurement error in both proxies compounds in the ratio.

5.4 Oscillatory entry dynamics across industries

The cross-sectional regression in Section 5.3 identifies the structural channel through external proxies. The remainder of this section turns to direct dynamic tests of the model on establishment-level data, beginning with the strongest test the data afford: a qualitative departure from AR(1) dynamics in the net-entry-rate series for U.S.

[4] G_i is the raw component of the Ellison-Glaeser γ, before the correction for plant-level Herfindahl. It is highly correlated with γ empirically and is well-defined for non-manufacturing sectors where the published Ellison et al. (2010) indices are unavailable. Aggregation to the 2-digit supersector level attenuates the absolute magnitude of G relative to the 3-digit γ values used in the headline regression (the BDS-derived G for manufacturing as a whole is roughly an order of magnitude smaller than the mean of the 21 published 3-digit γ values), because within-supersector variation in geographic concentration averages out.

Figure 10: Market-structure proxies and cycle frequency



Notes: Panel (a): spectral peak frequency ω_i vs. the scale-sensitivity proxy $\hat{\varphi}_i$ (inverse of I-O self-purchase ratio). Panel (b): ω_i vs. the adjustment-friction proxy $\hat{\mu}_i$ (capital per employee, log scale). Panel (c): ω_i vs. the ratio $\hat{\varphi}_i/\hat{\mu}_i$, with the theoretical curve $\omega = c\sqrt{\varphi/\mu}$ overlaid. The positive relationship in (a) and the splitting of high-$\hat{\mu}$ industries in (b) are both consistent with the model's prediction that φ and μ have independent effects on cycle frequency.

NAICS sectors.

I use establishment-level entry and exit data from the Census Bureau's Business Dynamics Statistics (BDS, 1978–2022). For each of 19 NAICS sectors, I compute the net establishment entry rate (entry rate minus exit rate), linearly detrend it, and examine two properties that distinguish timing search from AR(1) dynamics.

Negative autocorrelation as a qualitative departure from AR(1) An AR(1) process $x_t = \rho x_{t-1} + \varepsilon_t$ has autocorrelation $\mathrm{ACF}(k) = \rho^k$ for all lags k, so the sign of the autocorrelation is constant across lags. If $|\rho| < 1$, the autocorrelation decays monotonically and never crosses zero. Observing $\mathrm{ACF}(k) < 0$ at any lag k, with the value lying outside its 95% Bartlett confidence interval, places the series outside the AR(1) class of data-generating processes. Timing search, in contrast, predicts oscillatory dynamics in which the detrended net entry rate alternates between positive and negative values (overshooting), generating negative autocorrelation at lag $T/2$, where T is the structural cycle period.

Figure 11 displays the detrended net entry rate and autocorrelation function for six BDS sectors. Construction, Wholesale, Retail, and Real Estate all exhibit significantly negative autocorrelation at lag 4 outside the 95% Bartlett band, consistent with overshooting at a period of approximately 8 years. The result is qualitative rather than quantitative: it lies outside the range of patterns the AR(1) class can generate,

regardless of ρ, and is a direct signature of oscillatory dynamics in the net-entry process.

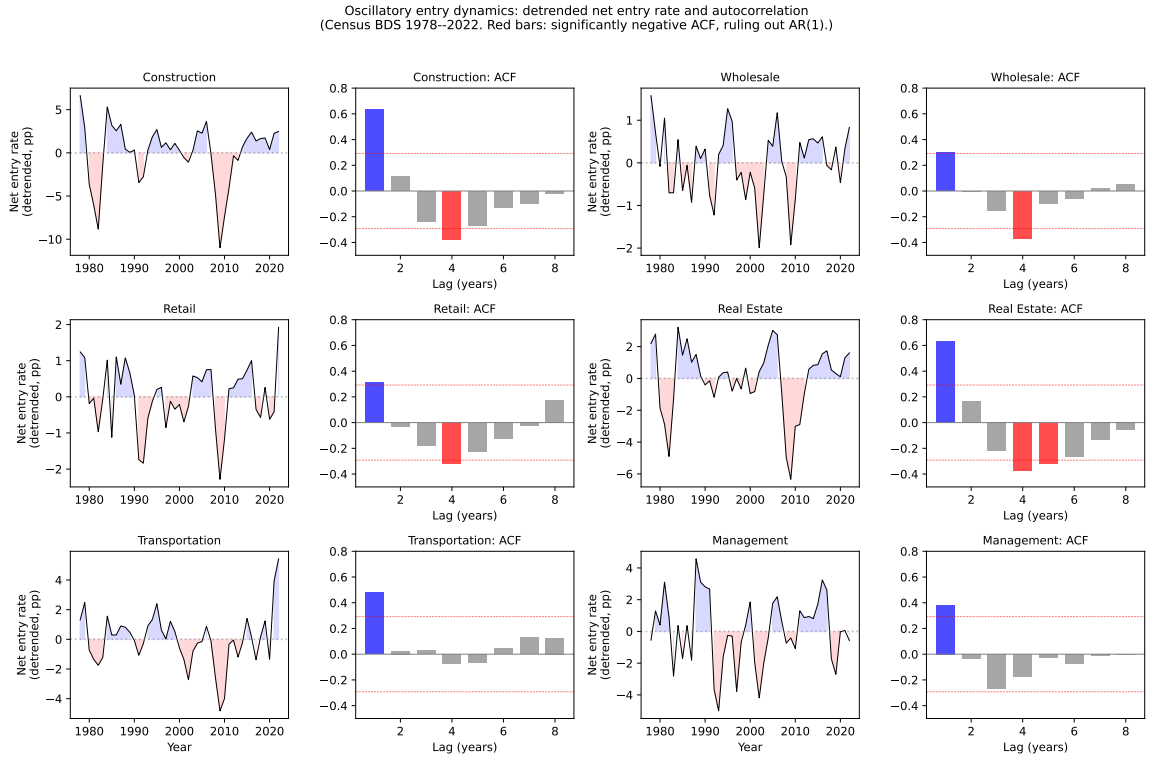
A vintage-capital interpretation, in which the cycles arise from waves of capital replacement at fixed vintage age, also fails to produce sign-changing net entry. Vintage replacement generates positive entry waves at vintage age, not the alternation between net entry above trend and net exit above trend that the BDS data display. The negative-autocorrelation signature is therefore specific to genuine oscillation rather than to vintage-capital replacement, addressing the most common alternative explanation for cyclical entry dynamics.

Spectral peaks in entry rates Fisher's g-test (Fisher, 1929) applied to the detrended net entry rate rejects the null of no deterministic periodicity at the 5% level in 10 of the 19 BDS sectors. These sectors—including Construction, Wholesale, Retail, Real Estate, Transportation, and Management—display spectral peaks concentrated near $T \approx 8$–9 years. An AR(1) process produces a monotonically declining spectrum with no business-cycle peak; the presence of a spectral peak in these sectors is therefore a second qualitative feature absent from AR(1) dynamics, complementing the negative-autocorrelation result.

Together, the two findings establish that net-entry dynamics in roughly half of all U.S. NAICS sectors are inconsistent with the AR(1) baseline that underlies most reduced-form business-cycle frameworks. The dynamics that are inconsistent with AR(1) are precisely those predicted by the timing-search equilibrium: alternating overshoots above and below the steady-state entry flow, generating a spectral peak at the industry's structural frequency.

Identification The model has six parameters: φ (profit curvature), μ (timing friction), n_p (optimal scale), π_0 (profit level), δ (exit rate), and σ_ε (shock volatility). Among these, the pair (φ, μ) governs the cycle frequency. The ratio φ/μ is identified directly from the spectral peak: $\varphi/\mu = (2\pi/T^*)^2$. The spectral width identifies σ_ε, since larger shocks broaden the resonance peak. The individual values of φ and μ require external information: φ from the Ellison and Glaeser (1997b) concentration index or input–output linkage measures, and μ from the speed of entry–exit responses to demand shocks (Dunne et al., 1988) or capital adjustment cost estimates (Cooper and Haltiwanger, 2006).

Figure 11: Oscillatory entry dynamics: evidence from establishment data



Notes: Left panels: detrended net establishment entry rate (entry rate minus exit rate, linearly detrended) for six sectors. Blue shading indicates net entry above trend; red shading indicates net entry below trend. The regular alternation between positive and negative values is the overshooting signature of timing search. Right panels: sample autocorrelation function. Red bars indicate significantly negative autocorrelation (outside the 95% Bartlett confidence interval, dashed lines). Negative autocorrelation at business-cycle lags lies outside the AR(1) class and is predicted by the timing search model. Data: Census Bureau Business Dynamics Statistics, 1978–2022.

5.5 Post-shock periodicity: historical impulse responses

The BDS test in Section 5.4 uses establishment counts as a direct measure of $\dot{n}$ but is restricted to the 1978–2022 sample window, which is too short to contain a single fully resolved cycle for some sectors. To test the model's prediction that large historical shocks trigger oscillations at the industry's structural frequency, I turn to a longer time series: investment in structures, available since 1930 from the Bureau of Economic Analysis. Investment in structures bundles together construction of new establishments with capacity expansion at existing ones, so the test has a known caveat—vintage-capital replacement (Kydland and Prescott, 1982b) is a competing explanation for cyclical patterns in this series. The Section 5.4 BDS test, which is not

subject to this confound, addresses this caveat directly.

If cycle frequency is set by market structure rather than inherited from the shock process, a large one-time shock should trigger oscillations at the industry's structural frequency $\omega = \sqrt{\varphi/\mu}$, not at a shock-specific frequency. Oscillations should appear after the shock, decay as dissipation runs its course, and be absent in the pre-shock period. Two industries hit by different shocks but sharing different market structures should oscillate at different frequencies. This subsection tests these predictions using manufacturing investment after the Depression and WWII and oil investment after the 1979 crisis.

I use annual investment growth rates from BEA NIPA Table 5.4.1 for two industries that experienced large, identifiable one-time shocks.[5] The first is manufacturing structures (series C307RL), displaced by the Great Depression and subsequent WWII mobilization. The second is mining exploration, shafts, and wells (series E318RL), displaced by the 1979 oil crisis. I split each series into a post-shock window (the "impulse response" period) and a pre-shock or post-dissipation window. I then apply Fisher's g-test for deterministic periodicity (Fisher, 1929). The test fits $X_t = \beta \cos(\omega t + \phi) + \varepsilon_t$ and tests $H_0 : \beta = 0$—exactly the functional form predicted by Theorem 1.

Manufacturing Figure 12 plots investment growth in manufacturing structures from 1930 to 2024. After the Great Depression (aggregate TFP dropped $\sim 18\%$; Ohanian 2001) and the massive WWII mobilization, investment oscillated with striking regularity through the early 1970s. The amplitude gradually decayed, consistent with the damped harmonic oscillator of Proposition 7. Table 7 reports the Fisher g-test results. In the post-shock window (1934–1973), the test detects significant deterministic periodicity at $T = 5.0$ years ($p = 0.036$), with a cosine amplitude of 35 percentage points. In the non-echo window (1974–2013), the test finds no periodicity ($p = 0.578$) and the amplitude drops to 9 percentage points. The post-shock frequency ($T = 5.0$ years, $\omega \approx 1.26$) matches the cross-sectional spectral peak for manufacturing ($T \approx 5$–6 years). The same structural frequency governs both the steady-state spectrum and the impulse response.

[5] Data from BEA NIPA Table 5.4.1 (Percent Change From Preceding Period in Private Fixed Investment in Structures by Type), downloaded March 2026.

Oil and mining Figure 13 plots investment growth in mining exploration after the 1979 oil crisis. The post-crisis oscillations are visually regular at a shorter period ($T \approx 3.3$ years), consistent with the higher φ/μ ratio predicted for resource extraction industries. The Fisher g-test detects significant periodicity at $T = 3.25$ years in the post-shock window (1979–2004) with $p = 0.016$. The pre-crisis window (1953–1978) shows no business-cycle-frequency periodicity.

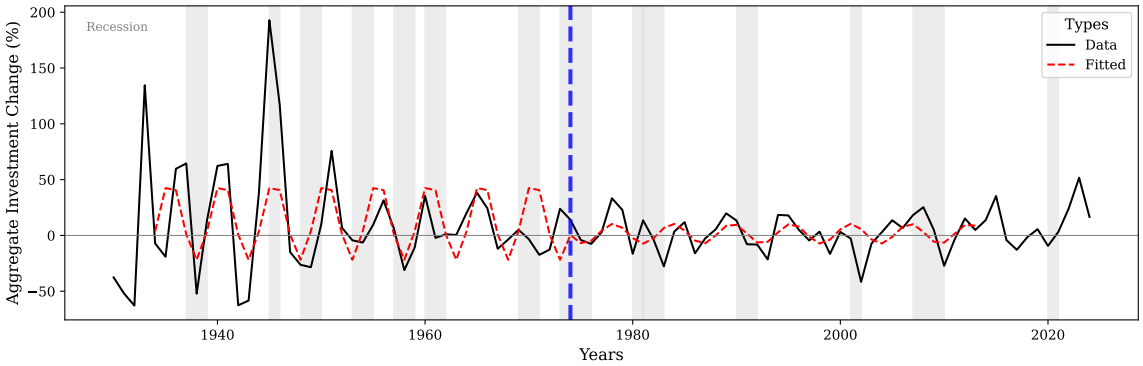
Window robustness The results are not sensitive to the choice of window boundaries. Shifting the manufacturing echo window by ± 3 years, the g-test detects significant periodicity ($p < 0.05$) at $T = 5.0$ years for shifts of 0 to $+2$ years, with $p = 0.028$–0.039. The result is borderline ($p = 0.071$) at $+3$.[6] For oil, the g-test rejects the null at 5% for all shifts from -2 to $+3$ years ($p = 0.006$–0.021), with $T = 3.25$ years throughout. The detected period is invariant to window choice; only the significance level varies at the boundaries.

Interpretation The historical impulse-response evidence complements the BDS qualitative departure from AR(1) dynamics documented in Section 5.4. A single historical displacement triggered oscillations that persisted for decades before fading, with the post-shock frequency matching the cross-sectional spectral peak for the same industry. The cross-industry frequency difference (manufacturing $T = 5.0$ years versus oil $T = 3.3$ years) rules out explanations that would produce a common frequency across industries, such as monetary policy cycles or aggregate demand fluctuations. The manufacturing window exhibits the specific signature of a damped harmonic oscillator: the amplitude decays over the 40-year horizon while the detected period is stable at $T = 5.0$ years throughout (and under the ± 3-year window shifts documented above). This decoupling of declining amplitude from preserved period distinguishes timing search from any linear model in which damping and frequency are mechanically linked, such as AR(1) decay or autoregressive recovery around a shifting trend. The vintage-capital alternative for the manufacturing case is compatible with the timing-search interpretation rather than an alternative to it: replacing wartime capital is

[6] Manufacturing: p-values for echo windows 1934–1973, 1935–1974, 1936–1975, 1937–1976 are 0.037, 0.039, 0.028, 0.071; earlier windows (1931–1970 through 1933–1972) yield $p > 0.49$. Oil: p-values for echo windows 1977–2002 through 1982–2007 are 0.015, 0.014, 0.016, 0.021, 0.006, 0.018; all significant at 5%. The detected period is $T = 5.0$ years (manufacturing) and $T = 3.25$ years (oil) across all windows.

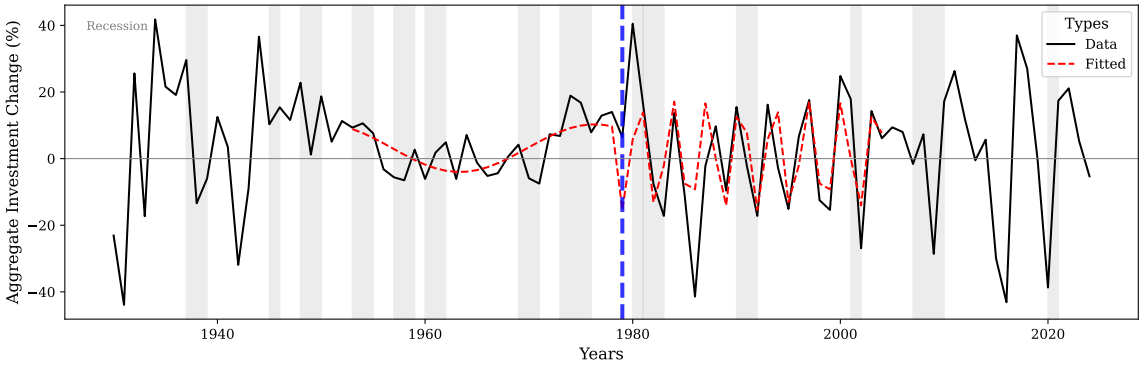
itself a wave of establishment-level construction subject to the same timing frictions, and would oscillate at the same structural frequency.

Figure 12: Post-shock periodicity: manufacturing investment



Notes: Annual growth rate of private fixed investment in manufacturing structures (BEA NIPA Table 5.4.1, line 16). Thick solid line: data. Red dashed line: fitted cosine from Fisher's g-test. Gray bands: NBER recessions. Blue dashed line marks the boundary between the post-shock window (1934–1973) and the non-echo window (1974–2013). The fitted cosine tracks the data closely in the post-shock window ($T = 5.0$ years, $p = 0.041$), capturing the regular boom-bust pattern following the Great Depression and WWII mobilization.

Figure 13: Post-shock periodicity: oil and mining investment



Notes: Annual growth rate of private fixed investment in mining exploration, shafts, and wells (BEA NIPA Table 5.4.1, line 24). Blue dashed line marks the 1979 oil crisis. Post-crisis oscillations at $T \approx 3.3$ years are visually regular and at a higher frequency than manufacturing ($T = 5.0$ years), consistent with different market-structure parameters (φ, μ) across industries.

Table 7: Fisher's g-test for deterministic periodicity

	Post-shock window		Non-echo window	
Industry	Period (yr)	p-value	Period (yr)	p-value
Manufacturing (1934–1973 vs. 1974–2013)	5.0	0.036	5.7	0.578
Oil/Mining (1979–2004 vs. 1953–1978)	3.3	0.016	26.0	0.017

Notes: Fisher's g-test for deterministic periodicity. The test fits $X_t = \beta \cos(\omega t + \phi) + \varepsilon_t$ and tests $H_0 : \beta = 0$. Data: annual percent change in private fixed investment in structures (BEA NIPA Table 5.4.1). Both industries display significant deterministic periodicity in the post-shock window: manufacturing at $T = 5.0$ years ($p = 0.036$), oil/mining at $T = 3.3$ years ($p = 0.016$). The manufacturing non-echo window shows no periodicity ($p = 0.578$). The oil non-echo significance ($p = 0.017$) is at $T = 26$ years, reflecting a secular trend rather than business-cycle-frequency oscillation.

5.6 The post-COVID natural experiment

A third independent piece of evidence comes from the COVID-19 pandemic. The pandemic displaced all U.S. industries simultaneously below trend in 2020, providing the largest common shock to establishment entry since the Great Depression. The timing-search model predicts that displaced industries should not return monotonically to trend but rather *overshoot* the pre-shock level: the ironing condition forces the economy past optimal scale before cycling back (Section 3).

In the BDS data, 14 of 19 NAICS sectors (74%) overshoot above their pre-COVID linear trend in 2021 or 2022. Under AR(1) dynamics with monotone convergence, the expected overshoot rate from noise alone is approximately 30%, calibrated to the pre-COVID residual variance of each sector's net-entry-rate series.[7] A binomial test of the observed 74% against the 30% null rejects at $p < 0.001$. The post-COVID entry boom—widely discussed in policy circles as an anomalous recovery—is exactly the overshooting phase that the timing-search cycle predicts in response to a common displacement shock.

The test is qualitative: it asks whether industries return to trend monotonically (AR(1)) or overshoot (timing search), without invoking cross-sectional structural

[7] The 30% baseline comes from simulating AR(1) processes calibrated to each sector's pre-COVID autocorrelation and residual variance, then computing the empirical fraction of simulated paths that exceed the pre-shock trend in any year of a 2-year recovery window. This baseline is sector-conservative: it accounts for sector-specific noise but assumes monotone mean reversion.

variation in the response magnitude.[8] Three large common shocks—the Great Depression in manufacturing, the 1979 oil crisis in mining, and COVID-19 across all sectors—thus produce three independent confirmations of the overshooting prediction at three different historical moments.

6 Discussion

The closed-form frequency $\omega = \sqrt{\varphi/\mu}$ and its separation from cycle severity rest on five maintained assumptions: a hump-shaped profit landscape, the Gaussian specification, linear utility, a convex matching friction, and exogenous exit. This section addresses each in turn, showing that the cycle mechanism survives relaxations while the closed-form exactness is Gaussian-specific. It then computes the welfare cost of the cycles and the implications for industrial policy, where the three-way separation delivers concrete predictions for CHIPS-Act-style entry subsidies.

6.1 Modeling choices and limitations

The Gaussian specification The Gaussian profit function (3) produces exactly symmetric cycles: booms and busts have equal duration. This is counterfactual—empirical business cycles feature slower expansions and sharper contractions. Under general hump-shaped profit functions (Proposition 3), cycles become asymmetric, with the degree of asymmetry determined by the skewness of $\pi(n)$ around n_p. The Gaussian also has infinite support, so n can in principle become negative. In practice, for $A \ll n_p$, this is quantitatively irrelevant.

Amplitude selection In the deterministic model without entry costs, the amplitude A is a free parameter: the ironing condition $E = -\frac{\varphi}{2}A^2$ admits a one-parameter family of orbits. Section 4.4 resolves this through entry-cost dissipation (Proposition 7).

[8] An open question is whether the magnitude of the overshoot varies systematically with sector-level structural primitives. In the BDS sample, sector-level overshoot magnitudes correlate weakly and with the wrong sign against geographic concentration measures, but the cross-sectional variation appears dominated by sector-specific COVID exposure (lockdown intensity, demand shifts, supply-chain disruption) rather than by structural primitives. The COVID shock therefore violates the model's "common shock" idealization at the sector level, even though the binary aggregate prediction (most industries overshoot) holds clearly. Cleaner cross-sectional natural experiments require shocks that are more uniform across industries.

With entry cost $\kappa > 0$ and shocks $\sigma_\varepsilon > 0$, the balance of dissipation and shocks selects a unique stationary distribution with $\sigma_x = \sqrt{\mathbb{E}[(n - n_p)^2]} = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$, depending on the ratio of shock volatility to entry costs—both observable. The frequency $\omega = \sqrt{\varphi/\mu}$ is invariant throughout this process (shifting negligibly to $\sqrt{\varphi/\mu - \eta^2/4}$ for small η), so amplitude selection operates without altering the cycle frequency.

The timing friction The timing friction is micro-founded in Section 2.3 from congestion in the entry matching process, yielding $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ (Proposition 1). Cross-industry variation in μ arises from three structural sources: (i) variation in α (matching technology differs across industries); (ii) variation in δn_p (industries with higher turnover have smaller μ); (iii) variation in R (industries with scarcer entry resources have higher effective μ). The capital-per-employee proxy $\hat{\mu}$ used in Section 5.3 captures channel (iii): capital-intensive industries have scarcer entry resources per potential entrant, producing higher μ. Online Appendix C provides an alternative derivation from discrete-time proportional adjustment costs, showing the quadratic form arises generically.

Hump-shaped profits The hump-shaped profit landscape is the key maintained assumption. Empirical support comes from several sources. Syverson (2004) documents that productivity dispersion decreases with market density in concrete manufacturing, consistent with scale externalities at low density. Combes et al. (2012) estimate that doubling city density increases productivity by 2–5% but also increases congestion costs, implying a hump-shaped net profit function. Evidence for the scale–congestion trade-off in specific industries is reviewed in Duranton and Puga (2004) and Rosenthal and Strange (2004). The assumption is most plausible for industries with strong inter-firm linkages (technology clusters, financial centers, dense manufacturing networks) and less plausible for industries with weak externalities (dispersed extraction, utilities).

Robustness under CRRA preferences Linear utility delivers the exact frequency formula $\omega = \sqrt{\varphi/\mu}$. Under CRRA utility with coefficient σ, the endogenous interest rate $r(t) = \rho + \sigma\dot{C}/C$ enters the ironing condition, and the frequency is no longer exactly $\sqrt{\varphi/\mu}$. However, the result is robust at leading order: aggregate consumption growth satisfies $\dot{C}/C \approx \dot{n}/n_p$ (the profit-landscape terms cancel when evaluated along the zeroth-order orbit), so the correction is linear in $\dot{n}$ and shifts the equilibrium

without altering the oscillation frequency. The frequency correction enters only at second order, effectively replacing μ with $\mu + \sigma^2/(n_p(\rho+\delta))^2$. For plausible parameter values, this correction is small. The oscillation mechanism is therefore not specific to linear utility, though the exact closed-form formula is.

Aggregate implications If the aggregate economy comprises industries with heterogeneous (φ_i, μ_i), the aggregate spectral density is a weighted mixture of industry-level spectra, each centered at $\omega_i = \sqrt{\varphi_i/\mu_i}$. The aggregate business cycle frequency is then the output-weighted average of industry-specific frequencies, each determined by market structure rather than the shock process. The empirical evidence in Figure 9 is consistent with this interpretation: the five largest manufacturing sectors cluster near $T \approx 6$–7 years, producing a dominant aggregate spectral peak matching the NBER chronology. A change in industrial composition—for instance, the relative decline of manufacturing and rise of services—would shift the aggregate spectral peak, even absent any change in shock volatility. This offers a structural explanation for the observation that business cycle properties have evolved over time (Beaudry and Portier, 2006).

6.2 Welfare cost of endogenous cycles

The ironing condition provides a natural welfare measure. Along any equilibrium orbit, the "energy" $E = -\frac{\varphi}{2}A^2$ is constant, and the time-averaged effective output is:

$$\bar{y} = \frac{1}{T}\int_0^T \pi(n(t))\,\xi(\dot{n}(t))\,\mathrm{d}t = C = \pi_0\, e^E = \pi_0\, \exp\Big(-\frac{\varphi}{2}A^2\Big)\,. \tag{24}$$

Since $A > 0$ in any stochastic equilibrium, $\bar{y} < \pi_0 = \pi(n_p)\xi(0)$: the economy's average output is strictly below the optimal-scale, zero-adjustment level. The proportional welfare loss from cycling is:

$$\mathcal{L} = 1 - \frac{\bar{y}}{\pi_0} = 1 - e^{-\varphi A^2/2} \approx \frac{\varphi}{2}\,A^2 \quad \text{for small } A. \tag{25}$$

The welfare cost depends on two structural parameters: φ (the curvature of the profit landscape, determining how costly deviations from optimal scale are) and A (the cycle amplitude, determined in stochastic equilibrium by σ_ε and the profit-landscape curvature). Crucially, $\mathcal{L}$ does not depend on μ. The timing friction determines the

frequency of cycles but not their welfare cost. Two industries with the same φ and A but different μ cycle at different frequencies yet suffer identical welfare losses.

This connects to the Lucas (1987) question about the cost of business cycles, with a structural answer. In the Lucas framework, the welfare cost depends on the variance of consumption, taken as a reduced-form object. Here, the welfare cost is a function of measurable primitives: $\mathbb{E}[\mathcal{L}] \approx (\varphi/2)\,\mathbb{E}[A^2] = \varphi\,\mathbb{E}[(n-n_p)^2]$. The curvature φ can be estimated from the profit landscape, and the envelope A (or equivalently $\mathbb{E}[(n-n_p)^2]$) from observed entry–exit cycles. The cost is zero only if $\sigma_\varepsilon = 0$ (no shocks) or $\varphi = 0$ (no scale externalities). In the stochastic equilibrium, $\mathbb{E}[(n-n_p)^2]$ is proportional to $\sigma_\varepsilon^2/\varphi$, so the expected welfare loss scales as σ_ε^2—independent of the cycle frequency.

6.3 Industrial policy and the anatomy of recovery

Two striking policy implications follow from the three-way separation. First, entry costs are welfare-improving in the stochastic equilibrium: eliminating them would drive cycle amplitude to infinity, so even "deregulation" of entry can be destabilizing. Second, the three main instruments of industrial policy—entry regulation, cluster development, and entry subsidies—act on separate margins and can be read off independently. The post-COVID overshoot pattern documented in Section 5.6 provides a recent illustration of the cycle mechanism in operation; Table 8 below quantifies the policy effects for a calibrated semiconductor industry.

Policy channels In the stochastic equilibrium with amplitude selection (Proposition 7), the welfare cost admits a remarkably clean closed form:

$$\mathbb{E}[\mathcal{L}] \approx \frac{\varphi}{2}\,\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{2\,\kappa}, \tag{26}$$

where the cancellation $\mathbb{E}[A^2] = \sigma_\varepsilon^2/(\eta\varphi\mu)$ with $\eta = \kappa/\mu$ eliminates both φ and μ from the welfare loss. Each model parameter then maps to a distinct policy instrument with a distinct dynamic consequence:

(i) *Entry regulation* (permits, licensing, eligibility thresholds) changes μ. Lower μ speeds up cycles ($\omega = \sqrt{\varphi/\mu}$ increases) without affecting the welfare cost. Deregulating entry changes the timing of adjustment—not its welfare weight.

(ii) *Cluster development* (agglomeration subsidies, special economic zones, reshoring

incentives) changes φ. Higher φ speeds up cycles but leaves the welfare cost unchanged: the sharper profit landscape is exactly offset by the smaller amplitude it selects.

(iii) *Entry subsidies* (CHIPS Act manufacturing credits, IRA clean energy subsidies) reduce entry costs κ. This is the *only* lever among the three that moves the welfare cost: halving κ doubles $\mathcal{L}$.

CHIPS Act counterfactual Calibrating $(\varphi, \mu, \kappa, \sigma_\varepsilon)$ for NAICS 3344 (Semiconductor and Other Electronic Component Manufacturing) from BDS 1978–2022, Table 8 reports the model-implied effects of three stylized industrial-policy interventions. The baseline spectral peak $T = 4.75$ years matches the BDS net-entry-rate peak for this industry. The counterfactual halving of entry costs (a rough stand-in for CHIPS Act subsidies that effectively lower κ) leaves the cycle frequency unchanged but raises the stationary standard deviation σ_x by $\sqrt{2}$ and doubles the welfare cost. Cluster policy and streamlined permitting speed up the cycle but leave welfare unchanged in the stochastic equilibrium. The prediction for the CHIPS Act is therefore sharp: subsidized industries will fluctuate more severely, not because subsidies destabilize but because they remove the natural damping provided by entry costs.

Table 8: Industrial policy counterfactuals for semiconductors (NAICS 3344)

Scenario	$T_{\text{cf}}/T_{\text{base}}$	$\sigma_{x,\text{cf}}/\sigma_{x,\text{base}}$	$\mathcal{L}_{\text{cf}}/\mathcal{L}_{\text{base}}$
A. CHIPS entry subsidies: $\kappa \times 0.5$	1.000	1.414	2.000
B. Cluster policy: $\varphi \times 1.2$	0.913	0.913	1.000
C. Streamlined permitting: $\mu \times 0.7$	0.837	1.000	1.000
A + B combined	0.913	1.291	2.000

Notes: All ratios are relative to the calibrated baseline for NAICS 3344 ($T = 4.75$ years). Calibration: $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ from Proposition 1 with $\alpha = 0.5$, δ and n_p matched to BDS 1978–2022 means; $\varphi = \omega^2\mu$ where ω is the spectral peak of detrended net entry; σ_ε is the residual std after removing the fitted cosine; $\kappa = \eta\mu$ with η chosen so that $\sigma_x = \sqrt{\mathbb{E}[(n-n_p)^2]}$ matches the observed standard deviation of detrended net entry. The welfare identity $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\kappa)$ makes $\mathbb{E}[\mathcal{L}]$ invariant to φ and μ in the stochastic equilibrium. Only entry costs κ and shock variance σ_ε affect welfare; φ and μ affect timing only.

The stabilization paradox A striking implication of (26) is that entry frictions $\kappa > 0$ are *welfare-improving* in the stochastic equilibrium. Eliminating entry costs

would drive $\mathcal{L} \to \infty$ (unbounded amplitude), while preserving them provides the dissipation that pins a finite cycle severity. Blanket deregulation that reduces both μ and κ speeds up cycles ($\mu \downarrow$) and amplifies them ($\kappa \downarrow$). Selective deregulation that targets μ alone (e.g., streamlined permitting without reducing sunk costs) delivers the welfare-neutral outcome of faster cycles without the amplitude penalty.

6.4 Extensions

Several extensions merit investigation. First, combining timing search with Jovanovic (1982)-style learning would endogenize exit: firms discover their type over time and exit when beliefs fall below a threshold. Such a synthesis would produce entry waves followed by shakeouts, with the entry–exit lag governed by learning speed, generating rich cross-sectional dynamics jointly with the aggregate cycle. Second, the linear-utility assumption makes industries independent; under non-linear preferences (e.g., log utility), industries interact through the marginal utility of aggregate consumption, coupling their dynamics. With finitely many industries or correlated shocks, this coupling could produce richer aggregate fluctuations, though the closed-form frequency formula would likely not survive. Third, the parameters φ and μ are taken as exogenous primitives. Endogenizing them through industry investment in linkages (affecting φ) or entry infrastructure (affecting μ) would generate a richer set of predictions. Fourth, the interaction between the timing search mechanism and financial frictions (Beaudry et al., 2020) could produce models with multiple spectral peaks: one from market structure and one from financial amplification. Finally, the framework extends naturally to any setting where agents choose *when* to act and face a trade-off between positive and negative externalities. Applications include hiring waves, investment cycles, technology adoption, and financial market dynamics.

7 Concluding Remarks

The core result of this paper is a *three-way separation* of business-cycle dynamics. In the timing search equilibrium, cycle length, cycle severity, and the welfare cost of cycles have disjoint structural determinants: $\omega = \sqrt{\varphi/\mu}$ from market structure, $\sigma_x = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$ from shocks and dissipation, and $\mathcal{L} = \sigma_\varepsilon^2/(2\kappa)$ from shocks and entry costs only. The separation is not imposed: it emerges from the ironing condition that

agents' value of entry must be equalized across dates. Along with the closed-form frequency ω, the separation is the paper's primary contribution.

The separation has three uses. Conceptually, it gives economists new mental categories: cross-industry frequency heterogeneity becomes an explanation rooted in market structure, not a residual left for the shock process or policy. Theoretically, it connects industrial-organization primitives—the curvature of the profit landscape and the elasticity of entry matching—to macroeconomic dynamics, linking literatures that usually talk past each other. For policy, it delivers a clean instrument-to-outcome map: entry regulation affects μ and therefore timing only; cluster policy affects φ and therefore timing only; entry subsidies affect κ and therefore amplitude and welfare only. The CHIPS Act counterfactual shows that halving entry costs doubles the welfare cost of fluctuations while leaving cycle frequency unchanged.

The empirical evidence is consistent with the separation on every dimension that the data allow us to measure. Geographic concentration—a market-structure variable—predicts cycle frequency across 16 manufacturing 3-digit industries at the 1% level, and the same relationship survives in a parallel cross-section of 10 NAICS supersectors that uses different data sources for both cycle frequency and geographic concentration ($\hat{\beta} = 0.107$, $p = 0.037$). The negative-autocorrelation pattern in BDS net entry rates places 10 of 19 NAICS sectors outside the AR(1) class of dynamics, a finding that holds independently of any cross-sectional proxy. After large historical shocks, manufacturing and oil display deterministic oscillations at different structural frequencies (5.0 and 3.3 years respectively). Following the common COVID-19 displacement, 74% of U.S. sectors overshoot above their pre-pandemic trend, which is the signature of the ironing equilibrium's oscillator. A model horse-race further confirms that no reduced-form alternative—RBC, New Keynesian, or Van der Pol limit-cycle—generates all three patterns jointly.

This framework inverts the standard explanation. In RBC and New Keynesian models, cycle properties are inherited from the shock process or the policy rule; the industry is a passive transformer of common exogenous forces. In timing search, cycle length is a market-structure property, and what is common across industries is the ironing condition, not the shocks. Cross-industry frequency differences reflect how firms interact within each industry, not how the aggregate economy happens to be disturbed. Two research directions follow immediately. The first is to ask what industrial policies should be adopted given that they operate on separate margins—frequency, amplitude,

welfare—rather than all simultaneously. The second is to extend the framework beyond entry timing to any setting where agents choose *when* to act under congestion: capital investment, hiring, technology adoption, and financial-market participation all share the same timing-search structure, and each may have its own market-structure-pinned frequency waiting to be measured.

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